Bazele inteligenței artificiale

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Laboratory contents

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3. Nonparametric models
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5. Dimensionality reduction
6. Clustering
7. Test 1
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10. Convolutional neural networks
11. Recurrent neural networks
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Content

- Attention pooling: Nadaraya-Watson kernel regression
- Attention scoring functions
- Bahdanau attention
- Multi-head attention
- Self-attention and positional encoding
- Transformer

1) Attention pooling: Nadaraya-Watson kernel regression

- it's a simple example for demonstrating the power of attention mechanism in ML

- **regression problem: given a dataset of input-output pairs $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$, how to learn f to predict the output $\hat{y} = f(x)$ for any new input x?**

- **solution**: write f as:

$$f(x) = \sum_{i=1}^{n} \underbrace{\alpha(x, x_i)}_{\text{attention weight}} y_i$$

query

key

value

- **question**: how do we choose the attention weight α ?
- **answer**: the sky is the limit ☺

- some examples (**Nadaraya-Watson** attention models):

$$f(x) = \sum_{i=1}^{n} \text{softmax}\left(-\frac{1}{2}(x - x_i)^2\right) y_i$$

non-parametric model: we don't have any weights or biases to learn

$$f(x) = \sum_{i=1}^{n} \text{softmax}\left(-\frac{1}{2}((x - x_i)\mathbf{w}\right)^2)y_i$$

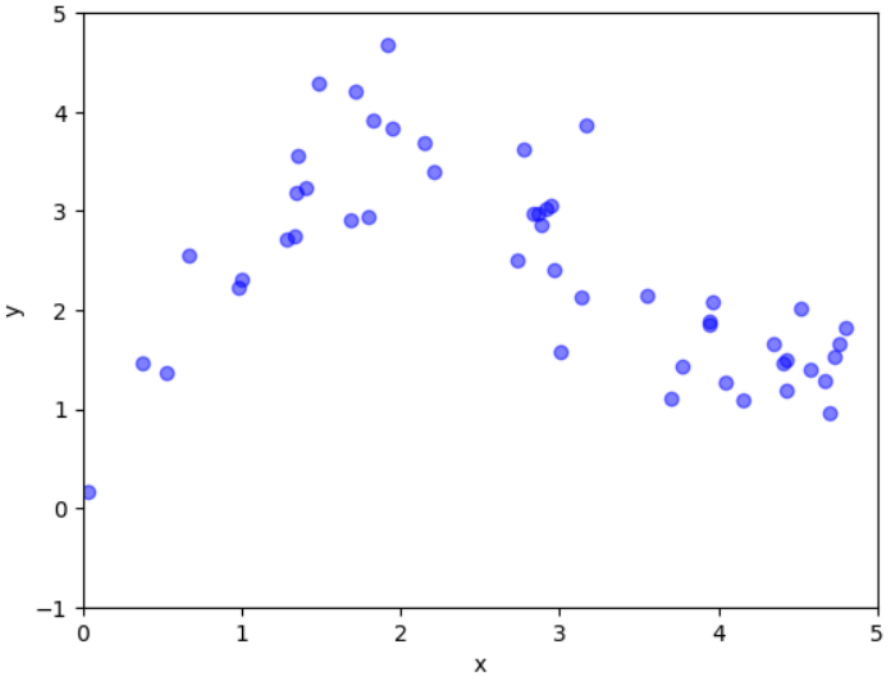
parametric model: we have **w** which is a learnable parameter

Why this formula with softmax? We will explain it in the next slides ☺

- we generate an artificial dataset according to the following nonlinear function, with the noise term ϵ:

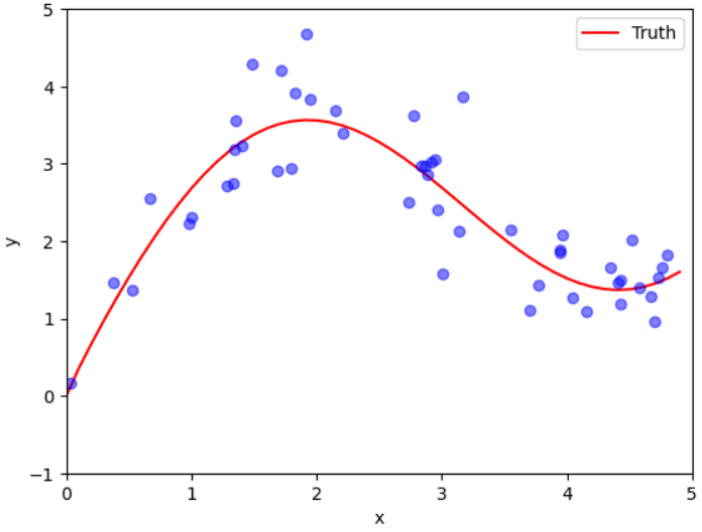
$$y_i = 2\sin(x_i) + x_i^{0.8} + \epsilon$$

```python
n_train = 50  # No. of training examples
x_train, _ = torch.sort(torch.rand(n_train) * 5)   # Training inputs

def f(x):
    return 2 * torch.sin(x) + x**0.8

y_train = f(x_train) + torch.normal(0.0, 0.5, (n_train,))  # Training outputs
```



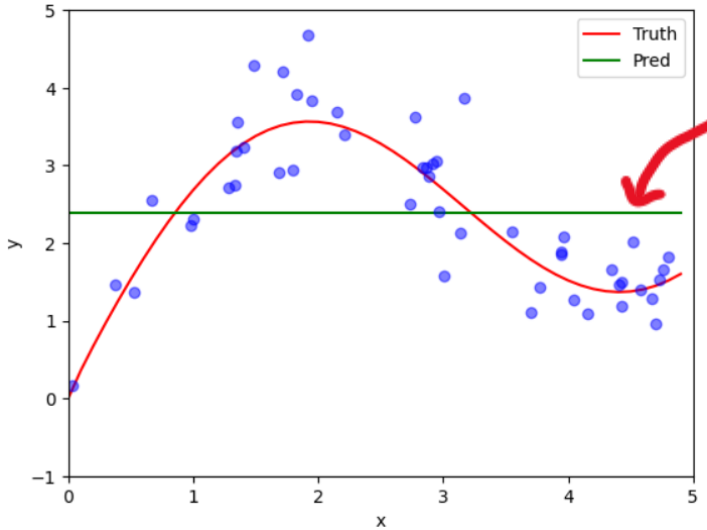
we generated 50 input-output pairs $\{(x_1, y_1), (x_2, y_2), \ldots, (x_{50}, y_{50})\}$

- our goal: we want to learn to estimate the blue line:



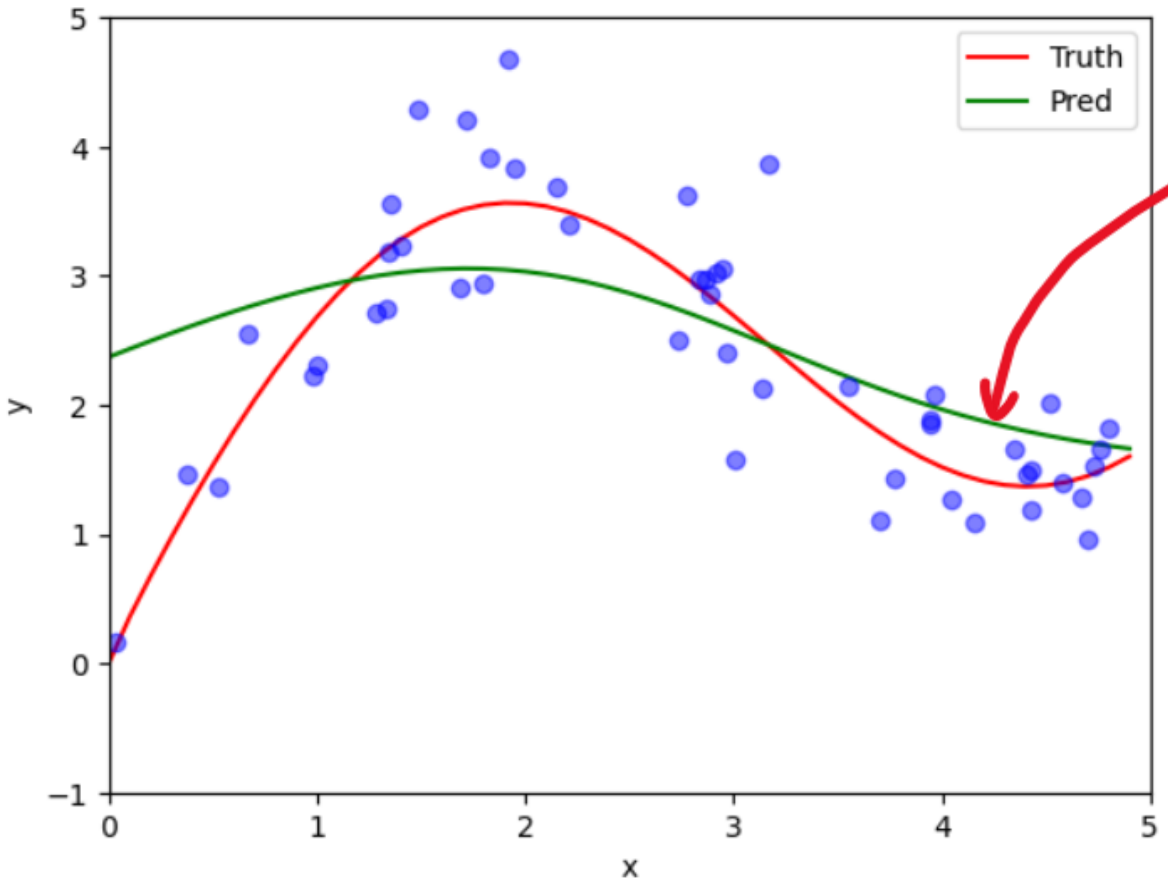
- one (stupid) solution:

$$f(x) = \frac{1}{n}\sum_{i=1}^{n} y_i$$



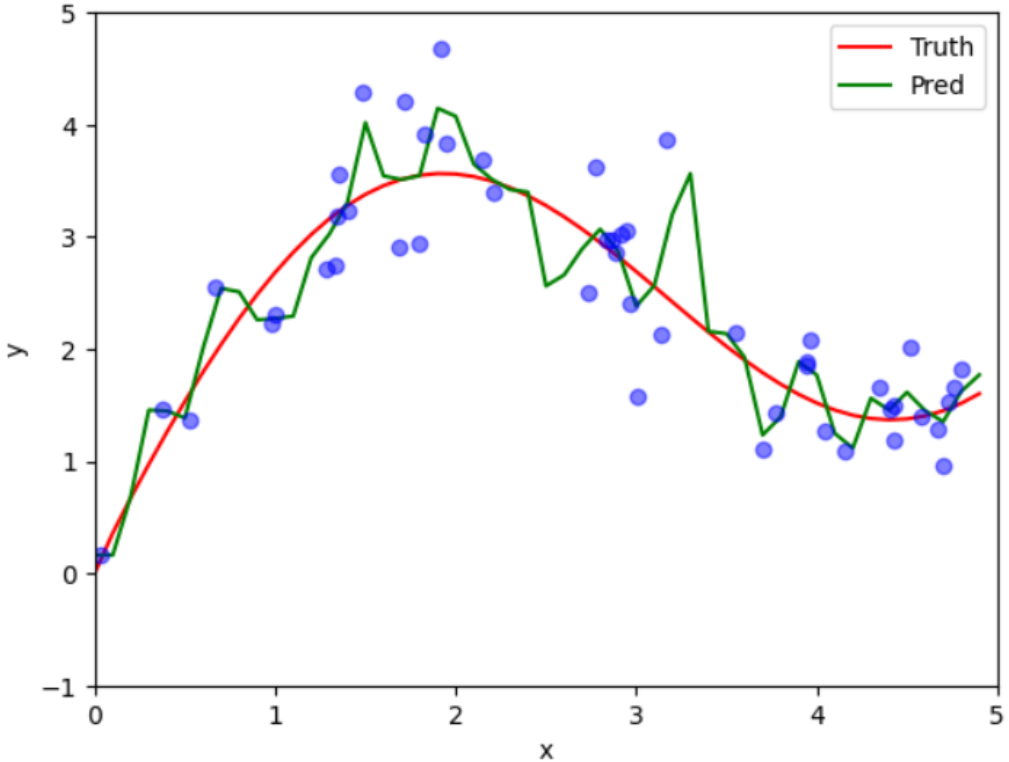
- one (smart) solution:

$$f(x) = \sum_{i=1}^{n} \mathrm{softmax}\left(-\frac{1}{2}(x - x_i)^2\right) y_i$$

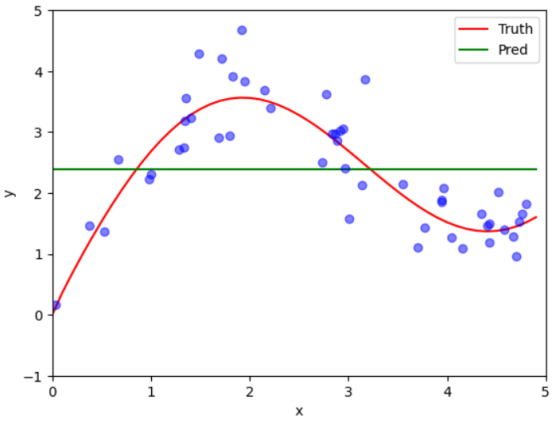


- one (genius) solution: $$f(x) = \sum_{i=1}^{n} \text{softmax}\left(-\frac{1}{2}((x - x_i)w)^2\right) y_i$$ we need to train this model to find ***w***

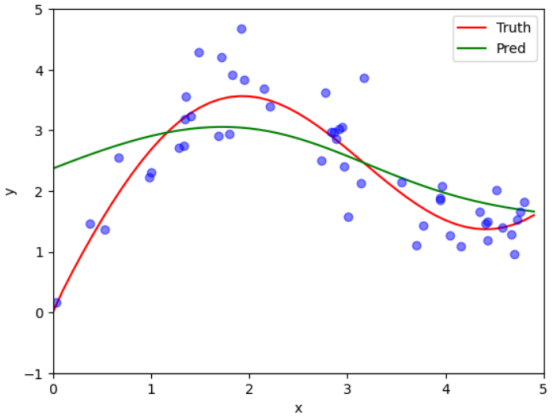
after training and finding the best ***w*** → **you have the *NWKernelRergession* class and the training loop already done in lab**



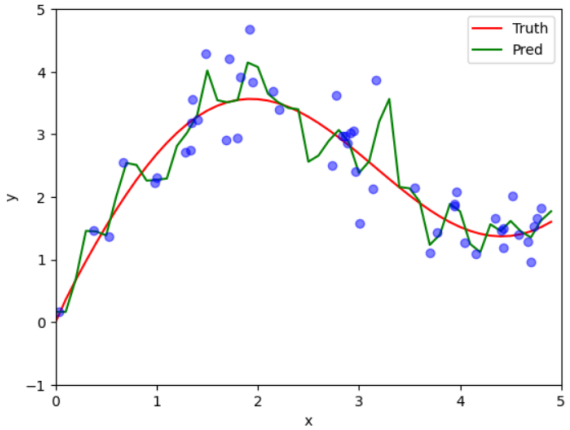
$$f(x) = \frac{1}{n}\sum_{i=1}^{n} y_i$$



$$f(x) = \sum_{i=1}^{n} \text{softmax}\left(-\frac{1}{2}(x - x_i)^2\right) y_i$$



$$f(x) = \sum_{i=1}^{n} \text{softmax}\left(-\frac{1}{2}((x - x_i)w)^2\right) y_i$$



2) Attention scoring functions

remember this?
(slide 5)

$$f(x) = \sum_{i=1}^{n} \underbrace{\alpha(x, x_i)}_{\text{attention weight}} y_i$$

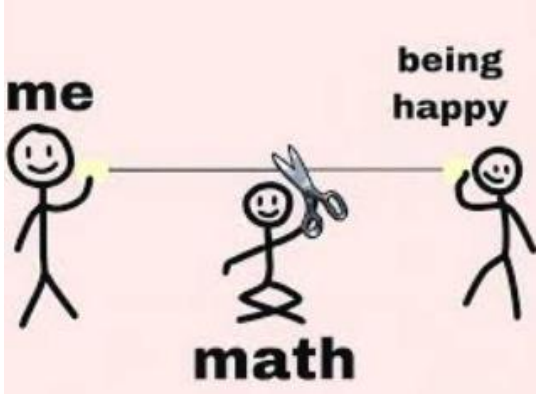
query

key

value

let's write it more mathematically:

$$\begin{cases} f(\mathbf{q}, (\mathbf{k_1}, \mathbf{v_1}), \ldots, (\mathbf{k_m}, \mathbf{v_m})) = \sum_{i=1}^{m} \alpha(\mathbf{q}, \mathbf{k_i})\mathbf{v_i} \in \mathbb{R}^v \\ \mathbf{q} \in \mathbb{R}^q \\ \mathbf{k_i} \in \mathbb{R}^k \\ \mathbf{v_i} \in \mathbb{R}^v \\ \alpha(\mathbf{q}, \mathbf{k_i}) \text{ - a scalar (a number ☺)} \end{cases}$$



$$\alpha(\mathbf{q}, \mathbf{k_i}) = softmax(a(\mathbf{q}, \mathbf{k_i})) = \frac{exp(a(\mathbf{q}, \mathbf{k_i}))}{\sum_{j=1}^{m} exp(a(\mathbf{q}, \mathbf{k_i}))} \in \mathbb{R}$$

scoring function

scoring functions:

learnable weights

$$a(\mathbf{q}, \mathbf{k}) = \mathbf{w_v}^T tanh(\mathbf{W_q}\mathbf{q} + \mathbf{W_k}\mathbf{k}) \in \mathbb{R}$$

- additive attention

$$a(\mathbf{q}, \mathbf{k}) = \frac{\mathbf{q}^T\mathbf{k}}{\sqrt{d}}$$

- scaled-dot product attention

len(q)

We have the classes for implementing these scoring functions in lab but we will explain them later with an example:

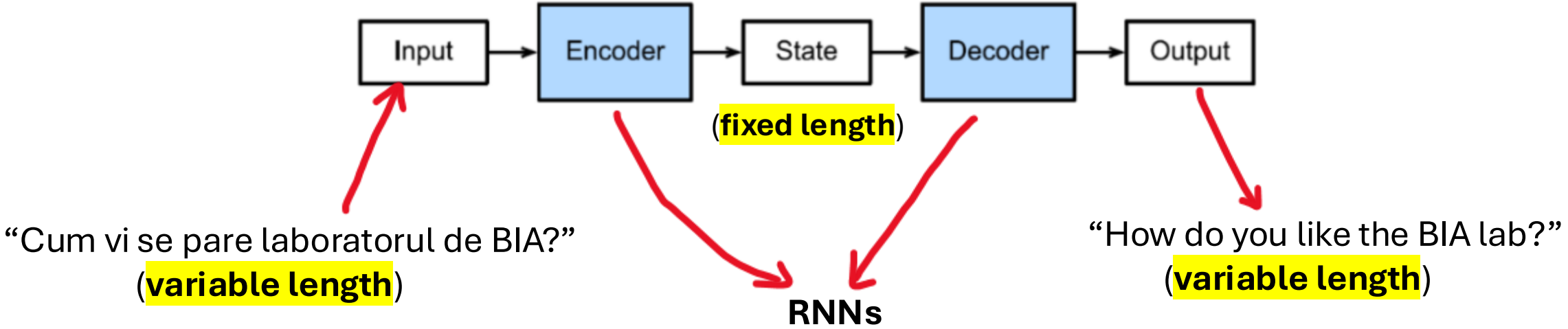
```python
class AdditiveAttention(nn.Module):
    """Additive attention."""
    def __init__(self, key_size, query_size, num_hiddens, dropout, **kwargs):
        super(AdditiveAttention, self).__init__(**kwargs)
        self.W_k = nn.Linear(key_size, num_hiddens, bias=False)
        self.W_q = nn.Linear(query_size, num_hiddens, bias=False)
        self.w_v = nn.Linear(num_hiddens, 1, bias=False)
        self.dropout = nn.Dropout(dropout)

    def forward(self, queries, keys, values, valid_lens):
        queries, keys = self.W_q(queries), self.W_k(keys)
        # After dimension expansion, shape of `queries` is: (`batch_size`, no.
        # of queries, 1, `num_hiddens`) and shape of `keys` is: (`batch_size`,
        # 1, no. of key-value pairs, `num_hiddens`). Sum them up with
        # broadcasting
        features = queries.unsqueeze(2) + keys.unsqueeze(1)
        features = torch.tanh(features)
        # There is only one output of `self.w_v`, so we remove the last
        # one-dimensional entry from the shape. Shape of `scores` is:
        # (`batch_size`, no. of queries, no. of key-value pairs)
        scores = self.w_v(features).squeeze(-1)
        self.attention_weights = masked_softmax(scores, valid_lens)
        # Shape of `values` is: (`batch_size`, no. of key-value pairs, value
        # dimension)
        return torch.bmm(self.dropout(self.attention_weights), values)
```

$$a(\mathbf{q}, \mathbf{k}) = \mathbf{w_v}^T tanh(\mathbf{W_q q} + \mathbf{W_k k}) \in \mathbb{R}$$

```python
class DotProductAttention(nn.Module):
    """Scaled dot product attention."""
    def __init__(self, dropout, **kwargs):
        super(DotProductAttention, self).__init__(**kwargs)
        self.dropout = nn.Dropout(dropout)

    # Shape of `queries` is: (`batch_size`, no. of queries, `d`)
    # Shape of `keys` is: (`batch_size`, no. of key-value pairs, `d`)
    # Shape of `values` is: (`batch_size`, no. of key-value pairs, value
    # dimension)
    # Shape of `valid_lens`: (`batch_size`,) or (`batch_size`, no. of queries)
    def forward(self, queries, keys, values, valid_lens=None):
        d = queries.shape[-1]
        # Use `keys.transpose(1,2)` to swap the last two dimensions of `keys`
        scores = torch.bmm(queries, keys.transpose(1,2)) / math.sqrt(d)
        self.attention_weights = masked_softmax(scores, valid_lens)
        return torch.bmm(self.dropout(self.attention_weights), values)
```

$$a(\mathbf{q}, \mathbf{k}) = \frac{\mathbf{q}^T \mathbf{k}}{\sqrt{d}}$$

3) Bahdanau attention



Remember this from the previous lab? We used 2 RNNs as encoder-decoder model to translate from one language to other (see next 2 slides):

Model

- 2 RNNs: one for encoder and one for decoder:

```python
class Encoder(nn.Module):
    """The base encoder interface for the encoder-decoder architecture."""
    def __init__(self, **kwargs):
        super(Encoder, self).__init__(**kwargs)

    def forward(self, X, *args):
        raise NotImplementedError
```

```python
class Seq2SeqEncoder(Encoder):
    """The RNN encoder for sequence to sequence learning."""
    def __init__(self, vocab_size, embed_size, num_hiddens, num_layers,
                 dropout=0, **kwargs):
        super(Seq2SeqEncoder, self).__init__(**kwargs)
        # Embedding layer
        self.embedding = nn.Embedding(vocab_size, embed_size)
        self.rnn = nn.GRU(embed_size, num_hiddens, num_layers,
                          dropout=dropout)

    def forward(self, X, *args):
        # The output `X` shape: (`batch_size`, `num_steps`, `embed_size`)
        X = self.embedding(X)
        # In RNN models, the first axis corresponds to time steps
        X = X.permute(1, 0, 2)
        # When state is not mentioned, it defaults to zeros
        output, state = self.rnn(X)
        # `output` shape: (`num_steps`, `batch_size`, `num_hiddens`)
        # `state` shape: (`num_layers`, `batch_size`, `num_hiddens`)
        return output, state
```

```python
class Decoder(nn.Module):
    """The base decoder interface for the encoder-decoder architecture."""
    def __init__(self, **kwargs):
        super(Decoder, self).__init__(**kwargs)

    def init_state(self, enc_outputs, *args):
        raise NotImplementedError

    def forward(self, X, state):
        raise NotImplementedError
```

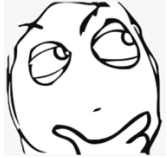
```python
class Seq2SeqDecoder(Decoder):
    """The RNN decoder for sequence to sequence learning."""
    def __init__(self, vocab_size, embed_size, num_hiddens, num_layers,
                 dropout=0, **kwargs):
        super(Seq2SeqDecoder, self).__init__(**kwargs)
        self.embedding = nn.Embedding(vocab_size, embed_size)
        self.rnn = nn.GRU(embed_size + num_hiddens, num_hiddens, num_layers,
                          dropout=dropout)
        self.dense = nn.Linear(num_hiddens, vocab_size)

    def init_state(self, enc_outputs, *args):
        return enc_outputs[1]

    def forward(self, X, state):
        # The output `X` shape: (`num_steps`, `batch_size`, `embed_size`)
        X = self.embedding(X).permute(1, 0, 2)
        # Broadcast `context` so it has the same `num_steps` as `X`
        context = state[-1].repeat(X.shape[0], 1, 1)
        X_and_context = torch.cat((X, context), 2)
        output, state = self.rnn(X_and_context, state)
        output = self.dense(output).permute(1, 0, 2)
        # `output` shape: (`batch_size`, `num_steps`, `vocab_size`)
        # `state` shape: (`num_layers`, `batch_size`, `num_hiddens`)
        return output, state
```

```python
class EncoderDecoder(nn.Module):
    """The base class for the encoder-decoder architecture."""
    def __init__(self, encoder, decoder, **kwargs):
        super(EncoderDecoder, self).__init__(**kwargs)
        self.encoder = encoder
        self.decoder = decoder

    def forward(self, enc_X, dec_X, *args):
        enc_outputs = self.encoder(enc_X, *args)
        dec_state = self.decoder.init_state(enc_outputs, *args)
        return self.decoder(dec_X, dec_state)
```

these are specified in the problem

```python
encoder = Seq2SeqEncoder(len(src_vocab), embed_size, num_hiddens, num_layers, dropout)
decoder = Seq2SeqDecoder(len(tgt_vocab), embed_size, num_hiddens, num_layers, dropout)
net = EncoderDecoder(encoder, decoder)
```

this is the model used for training in lab 10 ☺
we needed to use 5 classes to obtain it

- what is different now? 

- we use **Bahdanau attention** to redefine the decoder !


```python
class Encoder(nn.Module):
    """The base encoder interface for the encoder-decoder architecture."""
    def __init__(self, **kwargs):
        super(Encoder, self).__init__(**kwargs)

    def forward(self, X, *args):
        raise NotImplementedError
```

same as in lab 10

```python
class Seq2SeqEncoder(Encoder):
    """The RNN encoder for sequence to sequence learning."""
    def __init__(self, vocab_size, embed_size, num_hiddens, num_layers,
                 dropout=0, **kwargs):
        super(Seq2SeqEncoder, self).__init__(**kwargs)
        # Embedding layer
        self.embedding = nn.Embedding(vocab_size, embed_size)
        self.rnn = nn.GRU(embed_size, num_hiddens, num_layers,
                          dropout=dropout)

    def forward(self, X, *args):
        # The output `X` shape: (`batch_size`, `num_steps`, `embed_size`)
        X = self.embedding(X)
        # In RNN models, the first axis corresponds to time steps
        X = X.permute(1, 0, 2)
        # When state is not mentioned, it defaults to zeros
        output, state = self.rnn(X)
        # `output` shape: (`num_steps`, `batch_size`, `num_hiddens`)
        # `state` shape: (`num_layers`, `batch_size`, `num_hiddens`)
        return output, state
```

same as in lab 10

```python
class Decoder(nn.Module):
    """The base decoder interface for the encoder-decoder architecture."""
    def __init__(self, **kwargs):
        super(Decoder, self).__init__(**kwargs)

    def init_state(self, enc_outputs, *args):
        raise NotImplementedError

    def forward(self, X, state):
        raise NotImplementedError
```

same as in lab 10

```python
class AttentionDecoder(Decoder):
    """The base attention-based decoder interface."""
    def __init__(self, **kwargs):
        super(AttentionDecoder, self).__init__(**kwargs)

    @property
    def attention_weights(self):
        raise NotImplementedError
```

this is new !

```python
class Seq2SeqAttentionDecoder(AttentionDecoder):
    def __init__(self, vocab_size, embed_size, num_hiddens, num_layers,
                 dropout=0, **kwargs):
        super(Seq2SeqAttentionDecoder, self).__init__(**kwargs)
        self.attention = AdditiveAttention(
            num_hiddens, num_hiddens, num_hiddens, dropout)
        self.embedding = nn.Embedding(vocab_size, embed_size)
        self.rnn = nn.GRU(
            embed_size + num_hiddens, num_hiddens, num_layers,
            dropout=dropout)
        self.dense = nn.Linear(num_hiddens, vocab_size)

    def init_state(self, enc_outputs, enc_valid_lens, *args):
        # Shape of `outputs` is: (`num_steps`, `batch_size`, `num_hiddens`).
        # Shape of `hidden_state[0]` is: (`num_layers`, `batch_size`,
        # `num_hiddens`)
        outputs, hidden_state = enc_outputs
        return (outputs.permute(1, 0, 2), hidden_state, enc_valid_lens)

    def forward(self, X, state):
        # Shape of `enc_outputs` is: (`batch_size`, `num_steps`, `num_hiddens`).
        # Shape of `hidden_state[0]` is: (`num_layers`, `batch_size`,
        # `num_hiddens`)
        enc_outputs, hidden_state, enc_valid_lens = state
        # Shape of the output `X` is: (`num_steps`, `batch_size`, `embed_size`)
        X = self.embedding(X).permute(1, 0, 2)
        outputs, self._attention_weights = [], []
        for x in X:
            # Shape of `query` is: (`batch_size`, 1, `num_hiddens`)
            query = torch.unsqueeze(hidden_state[-1], dim=1)
            # Shape of `context` is: (`batch_size`, 1, `num_hiddens`)
            context = self.attention(
                query, enc_outputs, enc_outputs, enc_valid_lens)
            # Concatenate on the feature dimension
            x = torch.cat((context, torch.unsqueeze(x, dim=1)), dim=-1)
            # Reshape `x` as (1, `batch_size`, `embed_size` + `num_hiddens`)
            out, hidden_state = self.rnn(x.permute(1, 0, 2), hidden_state)
            outputs.append(out)
            self._attention_weights.append(self.attention.attention_weights)
        # After fully-connected layer transformation, shape of `outputs` is:
        # (`num_steps`, `batch_size`, `vocab_size`)
        outputs = self.dense(torch.cat(outputs, dim=0))
        return outputs.permute(1, 0, 2), [enc_outputs, hidden_state,
                                          enc_valid_lens]

    @property
    def attention_weights(self):
        return self._attention_weights
```

this is new !

```python
class EncoderDecoder(nn.Module):
    """The base class for the encoder-decoder architecture."""
    def __init__(self, encoder, decoder, **kwargs):
        super(EncoderDecoder, self).__init__(**kwargs)
        self.encoder = encoder
        self.decoder = decoder

    def forward(self, enc_X, dec_X, *args):
        enc_outputs = self.encoder(enc_X, *args)
        dec_state = self.decoder.init_state(enc_outputs, *args)
        return self.decoder(dec_X, dec_state)
```

same as in lab 10

these are specified in the problem

```python
encoder = Seq2SeqEncoder(len(src_vocab), embed_size, num_hiddens, num_layers, dropout)
decoder = Seq2SeqAttentionDecoder(len(tgt_vocab), embed_size, num_hiddens, num_layers, dropout)
net = EncoderDecoder(encoder, decoder)
```

this is the model used for training ☺
we need to use 6 classes to obtain it

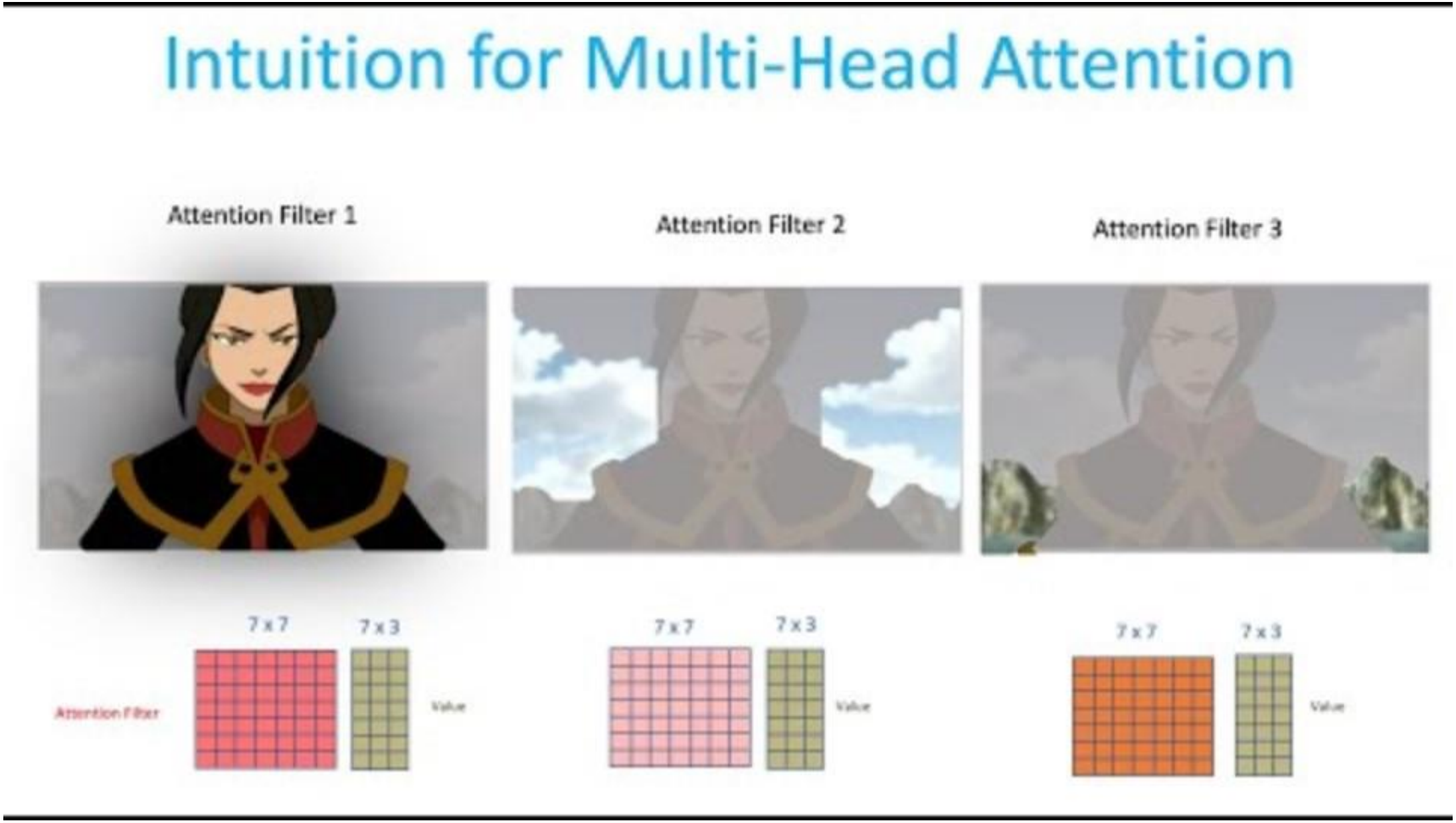
Now, before we start the Transformer part, let’s see some good tutorials…

Part 1

https://www.youtube.com/watch?v=dichIcUZfOw

Part 2

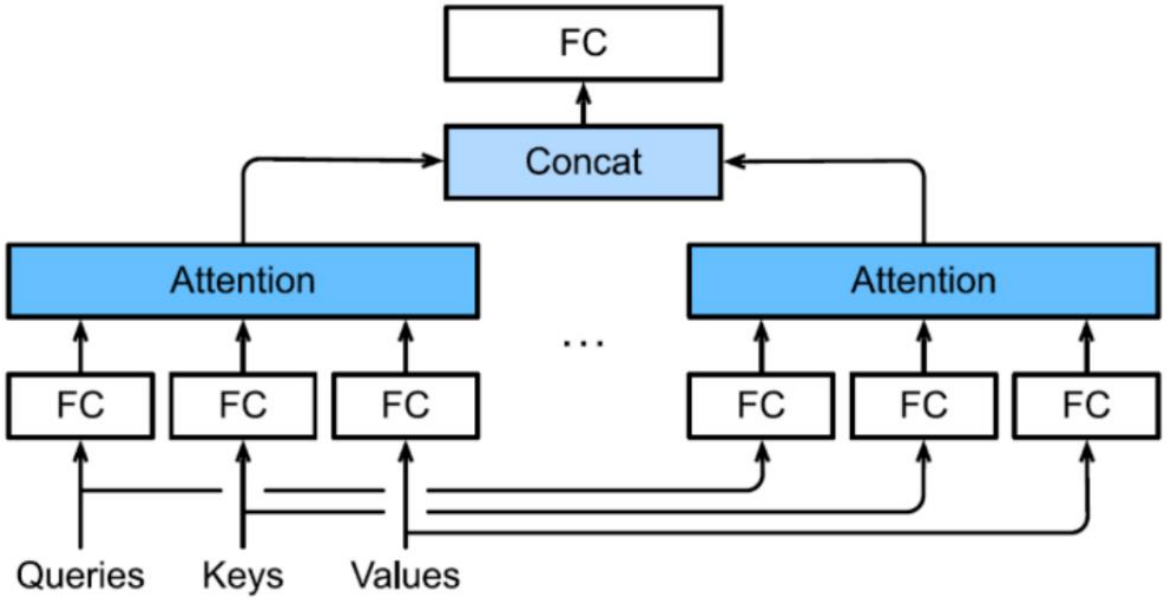
https://www.youtube.com/watch?v=mMa2PmYJlCo&t=14s



Part 3

https://www.youtube.com/watch?v=gJ9kaJsE78k&t=172s

4) Multi-head attention




```python
class MultiHeadAttention(nn.Module):
    """Multi-head attention."""
    def __init__(self, key_size, query_size, value_size, num_hiddens,
                 num_heads, dropout, bias=False, **kwargs):
        super(MultiHeadAttention, self).__init__(**kwargs)
        self.num_heads = num_heads
        self.attention = DotProductAttention(dropout)
        self.W_q = nn.Linear(query_size, num_hiddens, bias=bias)
        self.W_k = nn.Linear(key_size, num_hiddens, bias=bias)
        self.W_v = nn.Linear(value_size, num_hiddens, bias=bias)
        self.W_o = nn.Linear(num_hiddens, num_hiddens, bias=bias)

    def forward(self, queries, keys, values, valid_lens):
        # Shape of `queries`, `keys`, or `values` is:
        # (`batch_size`, no. of queries or key-value pairs, `num_hiddens`)
        # Shape of `valid_lens` is:
        # (`batch_size`,) or (`batch_size`, no. of queries)
        # After transposing, shape of output `queries`, `keys`, or `values` is:
        # (`batch_size` * `num_heads`, no. of queries or key-value pairs,
        # `num_hiddens` / `num_heads`)
        queries = transpose_qkv(self.W_q(queries), self.num_heads)
        keys = transpose_qkv(self.W_k(keys), self.num_heads)
        values = transpose_qkv(self.W_v(values), self.num_heads)

        if valid_lens is not None:
            # On axis 0, copy the first item (scalar or vector) for
            # `num_heads` times, then copy the next item, and so on
            valid_lens = torch.repeat_interleave(
                valid_lens, repeats=self.num_heads, dim=0)

        # Shape of `output` is: (`batch_size` * `num_heads`, no. of queries,
        # `num_hiddens` / `num_heads`)
        output = self.attention(queries, keys, values, valid_lens)

        # Shape of `output_concat` is:
        # (`batch_size`, no. of queries, `num_hiddens`)
        output_concat = transpose_output(output, self.num_heads)
        return self.W_o(output_concat)
```

5) Self-attention and positional encoding

- **self-attention**: queries, keys and values are from the same source:

$$\mathbf{y_i} = f(\mathbf{x_i}, (\mathbf{x_1}, \mathbf{x_1}), \ldots, (\mathbf{x_n}, \mathbf{x_n}))$$

$$p_{i,2j} = \sin\left(\frac{i}{10000^{2j/d}}\right), \quad p_{i,2j+1} = \cos\left(\frac{i}{10000^{2j/d}}\right)$$

```python
class PositionalEncoding(nn.Module):
    """Positional encoding."""
    def __init__(self, num_hiddens, dropout, max_len=1000):
        super(PositionalEncoding, self).__init__()
        self.dropout = nn.Dropout(dropout)
        # Create a long enough `P`
        self.P = torch.zeros((1, max_len, num_hiddens))
        X = torch.arange(max_len, dtype=torch.float32).reshape(
            -1, 1) / torch.pow(10000, torch.arange(
            0, num_hiddens, 2, dtype=torch.float32) / num_hiddens)
        self.P[:, :, 0::2] = torch.sin(X)
        self.P[:, :, 1::2] = torch.cos(X)

    def forward(self, X):
        X = X + self.P[:, :X.shape[1], :].to(X.device)
        return self.dropout(X)
```

https://kazemnejad.com/blog/transformer_architecture_positional_encoding/

6) Transformer

The Annotated Transformer

Attention is All You Need

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- *v2022: Austin Huang, Suraj Subramanian, Jonathan Sum, Khalid Almubarak, and Stella Biderman.*
- *Original: Sasha Rush.*

The Transformer has been on a lot of people's minds over the last ~~year~~ five years. This post presents an annotated version of the paper in the form of a line-by-line implementation. It reorders and deletes some sections from the original paper and adds comments throughout. This document itself is a working notebook, and should be a completely usable implementation. Code is available here.

Table of Contents

- Prelims
- Background

https://nlp.seas.harvard.edu/annotated-transformer/

Jay Alammar

Visualizing machine learning one concept at a time.

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Blog About

The Illustrated Transformer

Discussions: Hacker News (65 points, 4 comments), Reddit r/MachineLearning (29 points, 3 comments)

Translations: Arabic, Chinese (Simplified) 1, Chinese (Simplified) 2, French 1, French 2, Italian, Japanese, Korean, Persian, Russian, Spanish 1, Spanish 2, Vietnamese

Watch: MIT's Deep Learning State of the Art lecture referencing this post

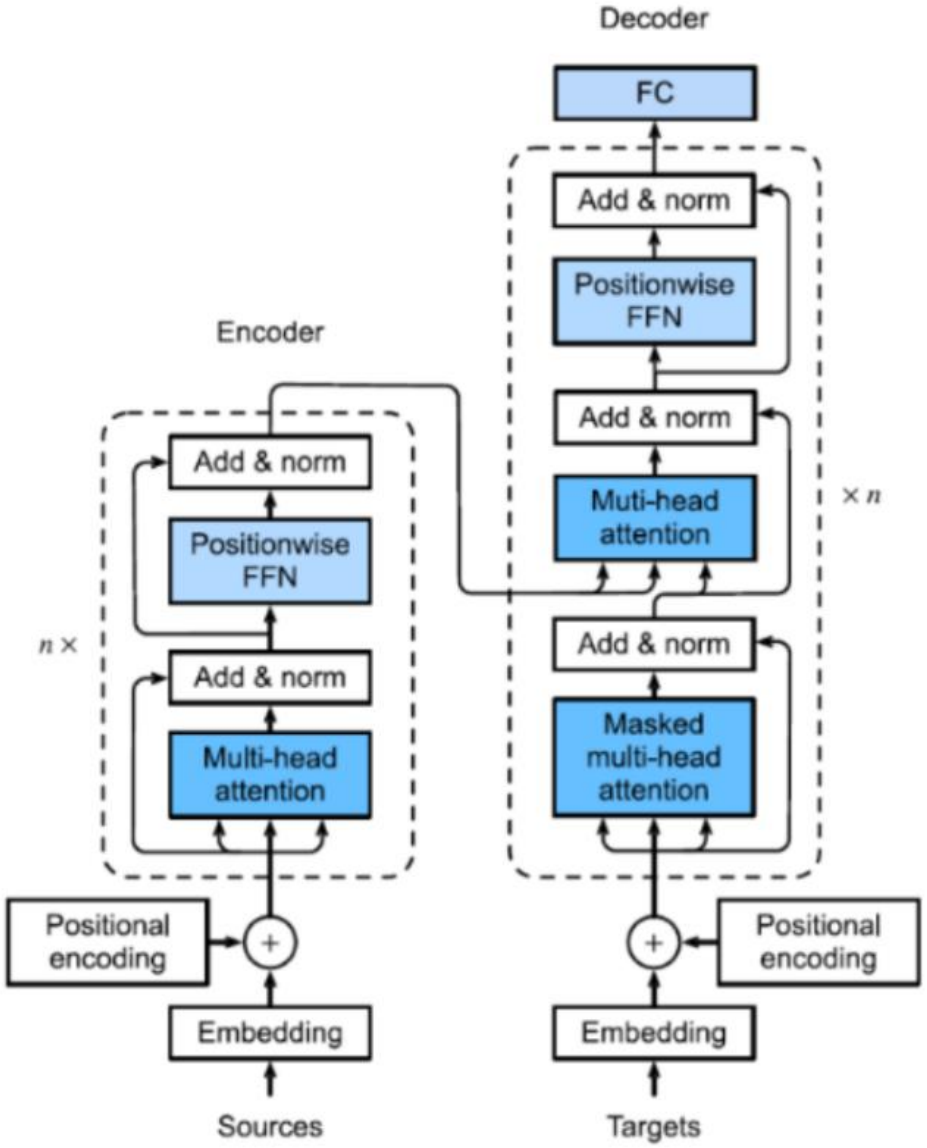
Featured in courses at Stanford, Harvard, MIT, Princeton, CMU and others

In the previous post, we looked at Attention – a ubiquitous method in modern deep learning models. Attention is a concept that helped improve the performance of neural machine translation applications. In this post, we will look at **The Transformer** – a model that uses attention to boost the speed with which these models can be trained. The Transformer outperforms the Google Neural Machine Translation model in specific tasks. The biggest benefit, however, comes from how The Transformer lends itself to parallelization. It is in fact Google Cloud's recommendation to use The Transformer as a reference model to use their Cloud TPU offering. So let's try to break the model apart and look at how it functions.

The Transformer was proposed in the paper Attention is All You Need. A TensorFlow implementation of it is available as a part of the Tensor2Tensor package. Harvard's NLP group created a guide annotating the paper with PyTorch implementation. In this post, we will attempt to oversimplify things a bit and introduce the concepts one by one to hopefully make it easier to understand to people without in-depth knowledge of the subject matter.

2020 Update: I've created a "Narrated Transformer" video which is a gentler approach to the topic:

https://jalammar.github.io/illustrated-transformer/



Attention Is All You Need

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arXiv:1706.03762v5 [cs.CL] 6 Dec 2017

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

1 Introduction

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

†Work performed while at Google Brain.

‡Work performed while at Google Research.

31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.

Ashish Vaswani

[PDF] **Attention is all you need**

A Vaswani - Advances in Neural Information Processing Systems, 2017 - user.phil.hhu.de

Attention is **all you need** **Attention** is **all you need** …

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Masking

```python
def sequence_mask(X, valid_len, value=0):
    """Mask irrelevant entries in sequences."""
    maxlen = X.size(1)
    mask = torch.arange((maxlen), dtype=torch.float32, device=X.device)[None, :] < valid_len[:, None]
    X[~mask] = value
    return X
```

+

```python
def masked_softmax(X, valid_lens):
    """Perform softmax operation by masking elements on the last axis."""
    # `X`: 3D tensor, `valid_lens`: 1D or 2D tensor
    if valid_lens is None:
        return nn.functional.softmax(X, dim=-1)
    else:
        shape = X.shape
        if valid_lens.dim() == 1:
            valid_lens = torch.repeat_interleave(valid_lens, shape[1])
        else:
            valid_lens = valid_lens.reshape(-1)
        # On the last axis, replace masked elements with a very large negative
        # value, whose exponentiation outputs 0
        X = sequence_mask(X.reshape(-1, shape[-1]), valid_lens,
                          value=-1e6)
        return F.softmax(X.reshape(shape), dim=-1)
```

Example

values beyond index 2 are all masked as zero

```
masked_softmax(torch.rand(2, 2, 4), torch.tensor([2, 3]))
```

```
tensor([[[0.4918, 0.5082, 0.0000, 0.0000],
         [0.4476, 0.5524, 0.0000, 0.0000]],

        [[0.4099, 0.1828, 0.4074, 0.0000],
         [0.3818, 0.2824, 0.3358, 0.0000]]])
```

values beyond index 3 are all masked as zero

values beyond indices 1 and 3 are all masked as zero

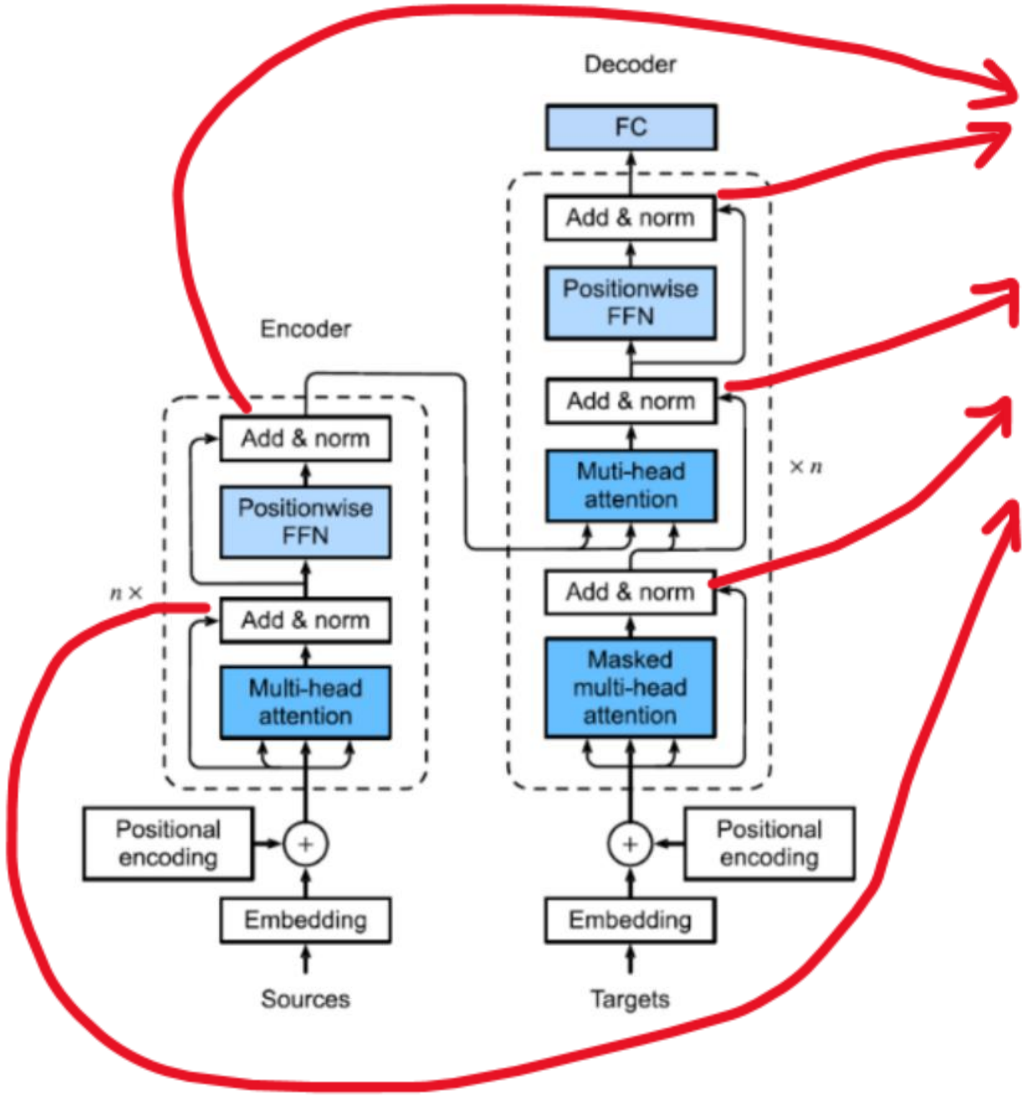
```
masked_softmax(torch.rand(2, 2, 4), torch.tensor([[1, 3], [2, 4]]))
```

```
tensor([[[1.0000, 0.0000, 0.0000, 0.0000],
         [0.3110, 0.3693, 0.3196, 0.0000]],

        [[0.4590, 0.5410, 0.0000, 0.0000],
         [0.2700, 0.1572, 0.4045, 0.1684]]])
```

values beyond indices 2 and 4 are all masked as zero

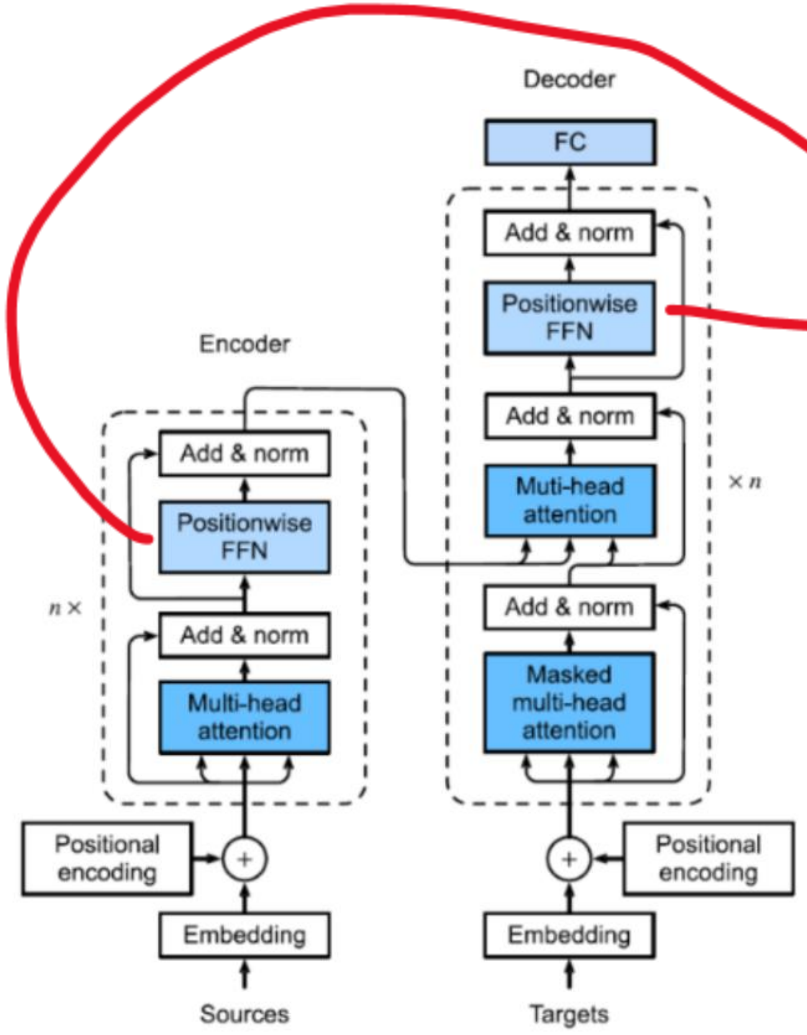
Create the Transformer model in code




```python
class AddNorm(nn.Module):
    """Residual connection followed by layer normalization."""
    def __init__(self, normalized_shape, dropout, **kwargs):
        super(AddNorm, self).__init__(**kwargs)
        self.dropout = nn.Dropout(dropout)
        self.ln = nn.LayerNorm(normalized_shape)

    def forward(self, X, Y):
        return self.ln(self.dropout(Y) + X)
```

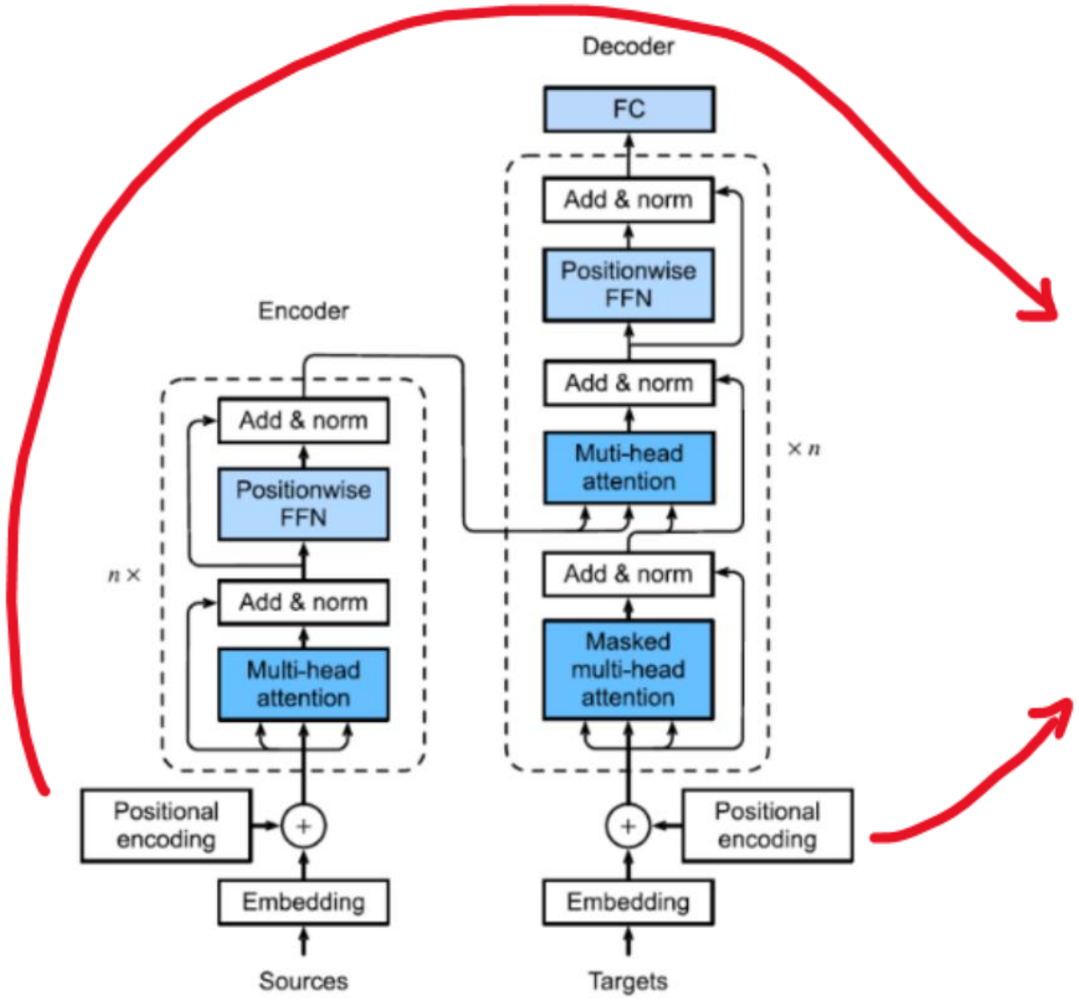
Create the Transformer model in code




```python
class PositionWiseFFN(nn.Module):
    """Position-wise feed-forward network."""
    def __init__(self, ffn_num_input, ffn_num_hiddens, ffn_num_outputs,
                 **kwargs):
        super(PositionWiseFFN, self).__init__(**kwargs)
        self.dense1 = nn.Linear(ffn_num_input, ffn_num_hiddens)
        self.relu = nn.ReLU()
        self.dense2 = nn.Linear(ffn_num_hiddens, ffn_num_outputs)

    def forward(self, X):
        return self.dense2(self.relu(self.dense1(X)))
```

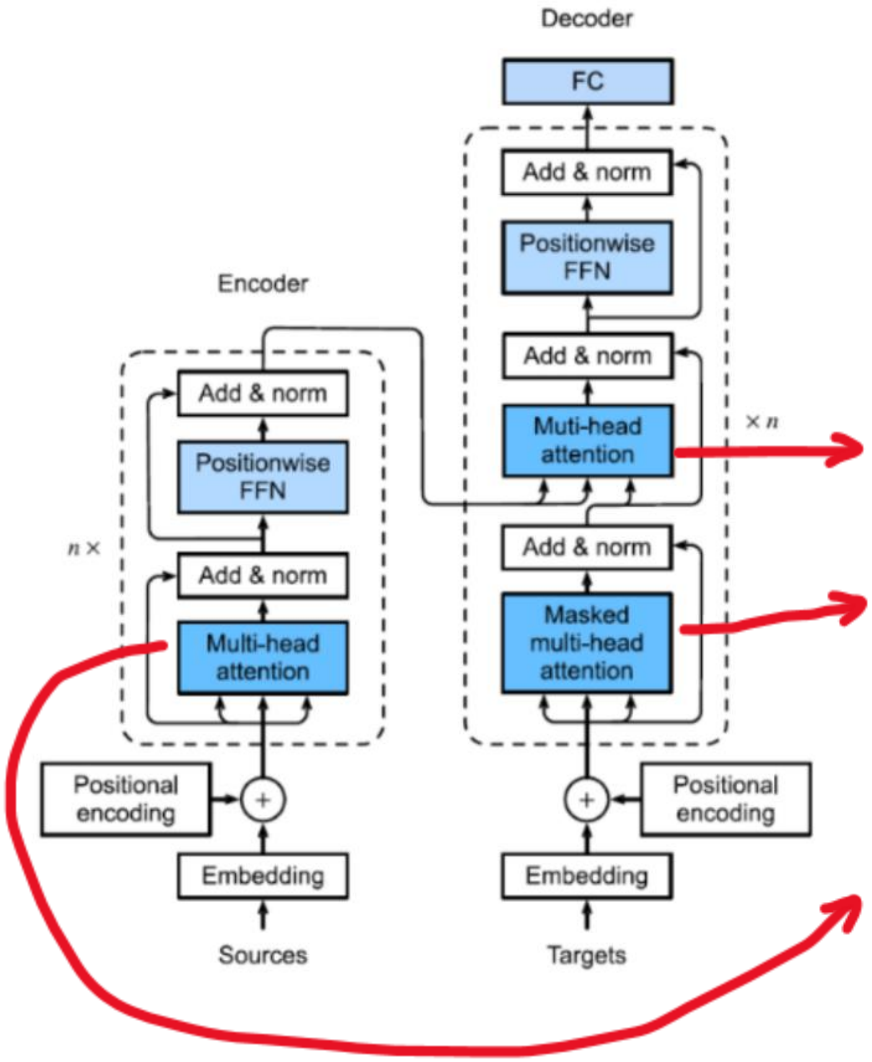
Create the Transformer model in code




```python
class PositionalEncoding(nn.Module):
    """Positional encoding."""
    def __init__(self, num_hiddens, dropout, max_len=1000):
        super(PositionalEncoding, self).__init__()
        self.dropout = nn.Dropout(dropout)
        # Create a long enough `P`
        self.P = torch.zeros((1, max_len, num_hiddens))
        X = torch.arange(max_len, dtype=torch.float32).reshape(
            -1, 1) / torch.pow(10000, torch.arange(
            0, num_hiddens, 2, dtype=torch.float32) / num_hiddens)
        self.P[:, :, 0::2] = torch.sin(X)
        self.P[:, :, 1::2] = torch.cos(X)

    def forward(self, X):
        X = X + self.P[:, :X.shape[1], :].to(X.device)
        return self.dropout(X)
```

Create the Transformer model in code




```python
class MultiHeadAttention(nn.Module):
    """Multi-head attention."""
    def __init__(self, key_size, query_size, value_size, num_hiddens,
                 num_heads, dropout, bias=False, **kwargs):
        super(MultiHeadAttention, self).__init__(**kwargs)
        self.num_heads = num_heads
        self.attention = DotProductAttention(dropout)
        self.W_q = nn.Linear(query_size, num_hiddens, bias=bias)
        self.W_k = nn.Linear(key_size, num_hiddens, bias=bias)
        self.W_v = nn.Linear(value_size, num_hiddens, bias=bias)
        self.W_o = nn.Linear(num_hiddens, num_hiddens, bias=bias)

    def forward(self, queries, keys, values, valid_lens):
        # Shape of `queries`, `keys`, or `values` is:
        # (`batch_size`, no. of queries or key-value pairs, `num_hiddens`)
        # Shape of `valid_lens` is:
        # (`batch_size`,) or (`batch_size`, no. of queries)
        # After transposing, shape of output `queries`, `keys`, or `values` is:
        # (`batch_size` * `num_heads`, no. of queries or key-value pairs,
        # `num_hiddens` / `num_heads`)
        queries = transpose_qkv(self.W_q(queries), self.num_heads)
        keys = transpose_qkv(self.W_k(keys), self.num_heads)
        values = transpose_qkv(self.W_v(values), self.num_heads)

        if valid_lens is not None:
            # On axis 0, copy the first item (scalar or vector) for
            # `num_heads` times, then copy the next item, and so on
            valid_lens = torch.repeat_interleave(
                valid_lens, repeats=self.num_heads, dim=0)

        # Shape of `output` is: (`batch_size` * `num_heads`, no. of queries,
        # `num_hiddens` / `num_heads`)
        output = self.attention(queries, keys, values, valid_lens)

        # Shape of `output_concat` is:
        # (`batch_size`, no. of queries, `num_hiddens`)
        output_concat = transpose_output(output, self.num_heads)
        return self.W_o(output_concat)
```

```python
def transpose_output(X, num_heads):
    """Reverse the operation of `transpose_qkv`."""
    X = X.reshape(-1, num_heads, X.shape[1], X.shape[2])
    X = X.permute(0, 2, 1, 3)
    return X.reshape(X.shape[0], X.shape[1], -1)
```

```python
def transpose_qkv(X, num_heads):
    """Transposition for parallel computation of multiple attention heads."""
    # Shape of input `X` is:
    # (`batch_size`, no. of queries or key-value pairs, `num_hiddens`).
    # Shape of output `X` is:
    # (`batch_size`, no. of queries or key-value pairs, `num_heads`,
    # `num_hiddens` / `num_heads`)
    X = X.reshape(X.shape[0], X.shape[1], num_heads, -1)

    # Shape of output `X` is:
    # (`batch_size`, `num_heads`, no. of queries or key-value pairs,
    # `num_hiddens` / `num_heads`)
    X = X.permute(0, 2, 1, 3)

    # Shape of `output` is:
    # (`batch_size` * `num_heads`, no. of queries or key-value pairs,
    # `num_hiddens` / `num_heads`)
    return X.reshape(-1, X.shape[2], X.shape[3])
```

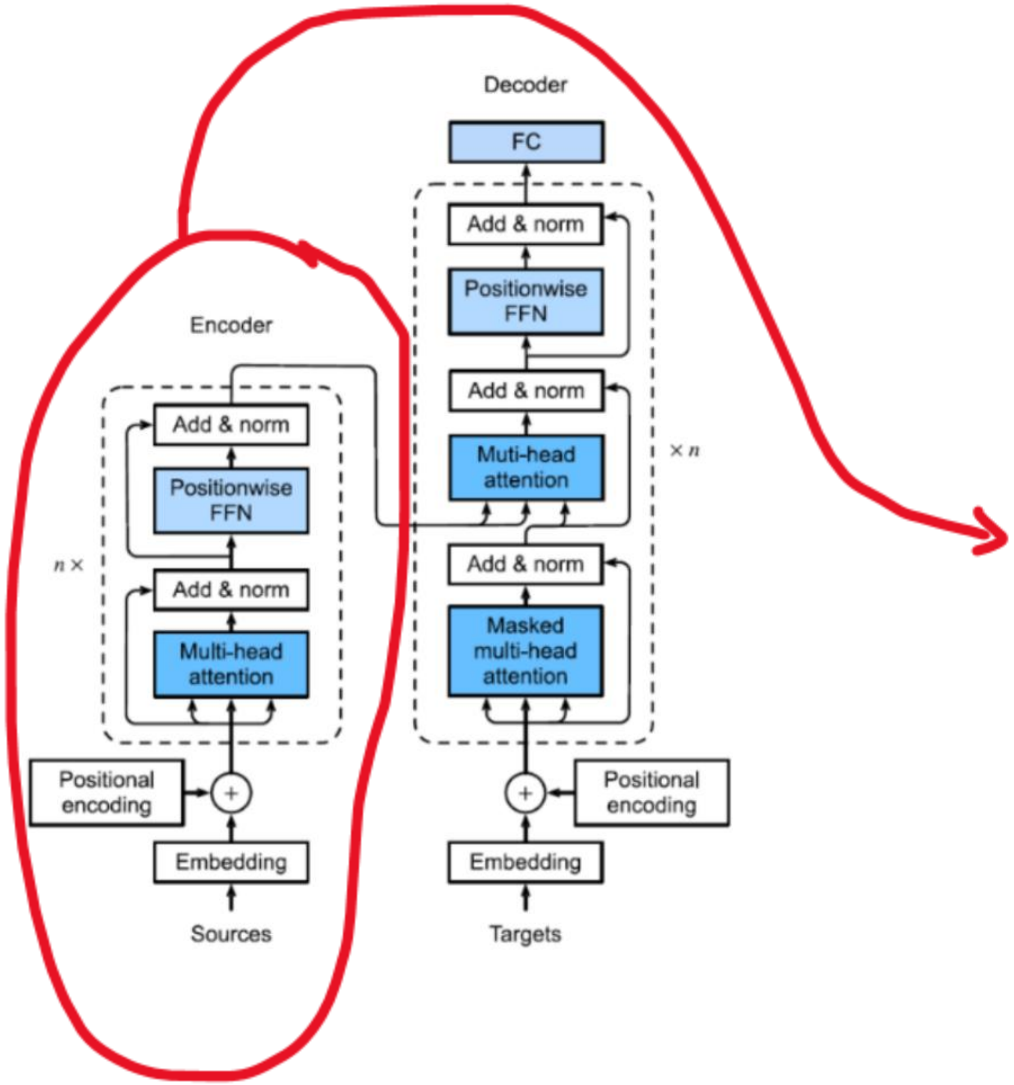
```python
class DotProductAttention(nn.Module):
    """Scaled dot product attention."""
    def __init__(self, dropout, **kwargs):
        super(DotProductAttention, self).__init__(**kwargs)
        self.dropout = nn.Dropout(dropout)

    # Shape of `queries` is: (`batch_size`, no. of queries, `d`)
    # Shape of `keys` is: (`batch_size`, no. of key-value pairs, `d`)
    # Shape of `values` is: (`batch_size`, no. of key-value pairs, value
    # dimension)
    # Shape of `valid_lens`: (`batch_size`,) or (`batch_size`, no. of queries)
    def forward(self, queries, keys, values, valid_lens=None):
        d = queries.shape[-1]
        # Use `keys.transpose(1,2)` to swap the last two dimensions of `keys`
        scores = torch.bmm(queries, keys.transpose(1,2)) / math.sqrt(d)
        self.attention_weights = masked_softmax(scores, valid_lens)
        return torch.bmm(self.dropout(self.attention_weights), values)
```

```python
def sequence_mask(X, valid_len, value=0):
    """Mask irrelevant entries in sequences."""
    maxlen = X.size(1)
    mask = torch.arange((maxlen), dtype=torch.float32, device=X.device)[None, :] < valid_len[:, None]
    X[~mask] = value
    return X
```

```python
def masked_softmax(X, valid_lens):
    """Perform softmax operation by masking elements on the last axis."""
    # `X`: 3D tensor, `valid_lens`: 1D or 2D tensor
    if valid_lens is None:
        return nn.functional.softmax(X, dim=-1)
    else:
        shape = X.shape
        if valid_lens.dim() == 1:
            valid_lens = torch.repeat_interleave(valid_lens, shape[1])
        else:
            valid_lens = valid_lens.reshape(-1)
        # On the last axis, replace masked elements with a very large negative
        # value, whose exponentiation outputs 0
        X = sequence_mask(X.reshape(-1, shape[-1]), valid_lens,
                          value=-1e6)
        return F.softmax(X.reshape(shape), dim=-1)
```

Create the Transformer model in code




```python
class Encoder(nn.Module):
    """The base encoder interface for the encoder-decoder architecture."""
    def __init__(self, **kwargs):
        super(Encoder, self).__init__(**kwargs)

    def forward(self, X, *args):
        raise NotImplementedError
```

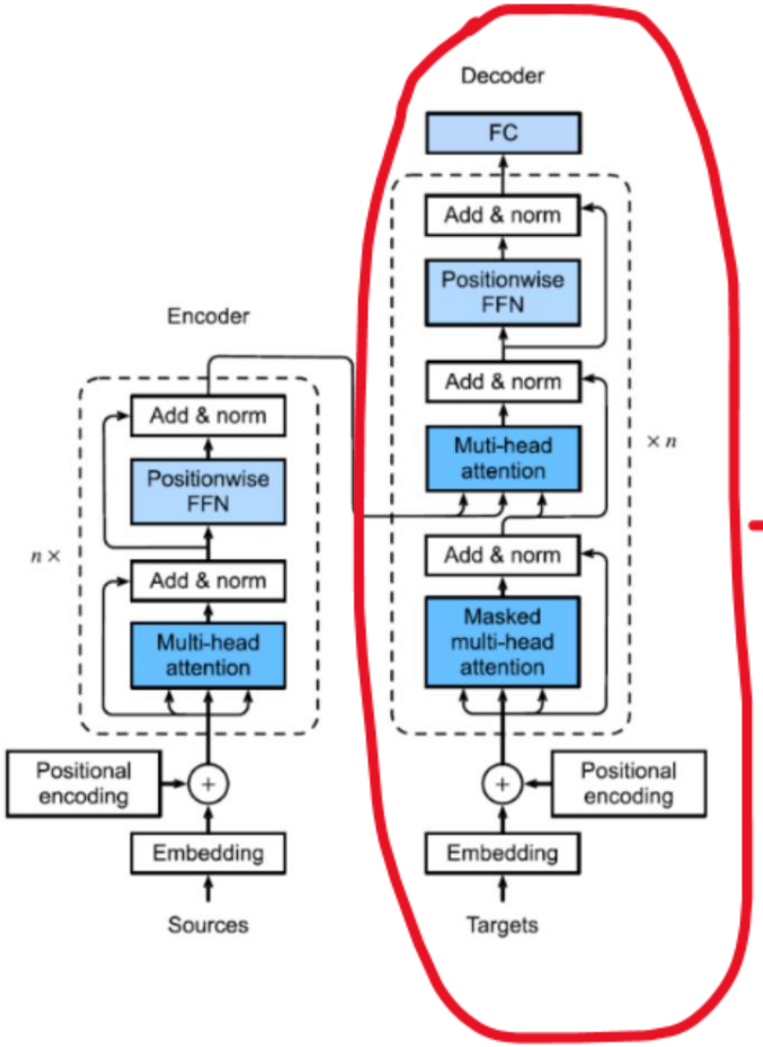
```python
class EncoderBlock(nn.Module):
    """Transformer encoder block."""
    def __init__(self, key_size, query_size, value_size, num_hiddens,
                 norm_shape, ffn_num_input, ffn_num_hiddens, num_heads,
                 dropout, use_bias=False, **kwargs):
        super(EncoderBlock, self).__init__(**kwargs)
        self.attention = MultiHeadAttention(
            key_size, query_size, value_size, num_hiddens, num_heads, dropout,
            use_bias)
        self.addnorm1 = AddNorm(norm_shape, dropout)
        self.ffn = PositionWiseFFN(
            ffn_num_input, ffn_num_hiddens, num_hiddens)
        self.addnorm2 = AddNorm(norm_shape, dropout)

    def forward(self, X, valid_lens):
        Y = self.addnorm1(X, self.attention(X, X, X, valid_lens))
        return self.addnorm2(Y, self.ffn(Y))
```

```python
class TransformerEncoder(Encoder):
    """Transformer encoder."""
    def __init__(self, vocab_size, key_size, query_size, value_size,
                 num_hiddens, norm_shape, ffn_num_input, ffn_num_hiddens,
                 num_heads, num_layers, dropout, use_bias=False, **kwargs):
        super(TransformerEncoder, self).__init__(**kwargs)
        self.num_hiddens = num_hiddens
        self.embedding = nn.Embedding(vocab_size, num_hiddens)
        self.pos_encoding = PositionalEncoding(num_hiddens, dropout)
        self.blks = nn.Sequential()
        for i in range(num_layers):
            self.blks.add_module("block"+str(i),
                EncoderBlock(key_size, query_size, value_size, num_hiddens,
                             norm_shape, ffn_num_input, ffn_num_hiddens,
                             num_heads, dropout, use_bias))

    def forward(self, X, valid_lens, *args):
        # Since positional encoding values are between -1 and 1, the embedding
        # values are multiplied by the square root of the embedding dimension
        # to rescale before they are summed up
        X = self.pos_encoding(self.embedding(X) * math.sqrt(self.num_hiddens))
        self.attention_weights = [None] * len(self.blks)
        for i, blk in enumerate(self.blks):
            X = blk(X, valid_lens)
            self.attention_weights[
                i] = blk.attention.attention.attention_weights
        return X
```

Create the Transformer model in code




```python
class Decoder(nn.Module):
    """The base decoder interface for the encoder-decoder architecture."""
    def __init__(self, **kwargs):
        super(Decoder, self).__init__(**kwargs)

    def init_state(self, enc_outputs, *args):
        raise NotImplementedError

    def forward(self, X, state):
        raise NotImplementedError

class AttentionDecoder(Decoder):
    """The base attention-based decoder interface."""
    def __init__(self, **kwargs):
        super(AttentionDecoder, self).__init__(**kwargs)

    @property
    def attention_weights(self):
        raise NotImplementedError
```

```python
class DecoderBlock(nn.Module):
    # The `i`-th block in the decoder
    def __init__(self, key_size, query_size, value_size, num_hiddens, norm_shape, ffn_num_input, ffn_num_hiddens, num_heads, dropout, i, **kwargs):
        super(DecoderBlock, self).__init__(**kwargs)
        self.i = i
        self.attention1 = MultiHeadAttention(key_size, query_size, value_size, num_hiddens, num_heads, dropout)
        self.addnorm1 = AddNorm(norm_shape, dropout)
        self.attention2 = MultiHeadAttention(key_size, query_size, value_size, num_hiddens, num_heads, dropout)
        self.addnorm2 = AddNorm(norm_shape, dropout)
        self.ffn = PositionWiseFFN(ffn_num_input, ffn_num_hiddens, num_hiddens)
        self.addnorm3 = AddNorm(norm_shape, dropout)

    def forward(self, X, state):
        enc_outputs, enc_valid_lens = state[0], state[1]
        # During training, all the tokens of any output sequence are processed at the same time, so `state[2][self.i]` is `None` as initialized.
        # When decoding any output sequence token by token during prediction, `state[2][self.i]` contains representations of the decoded output at
        # the `i`-th block up to the current time step
        if state[2][self.i] is None:
            key_values = X
        else:
            key_values = torch.cat((state[2][self.i], X), axis=1)
        state[2][self.i] = key_values
        if self.training:
            batch_size, num_steps, _ = X.shape
            # Shape of `dec_valid_lens` is: (`batch_size`, `num_steps`), where every row is [1, 2, ..., `num_steps`]
            dec_valid_lens = torch.arange(1, num_steps + 1, device=X.device).repeat(batch_size, 1)
        else:
            dec_valid_lens = None

        # Self-attention
        X2 = self.attention1(X, key_values, key_values, dec_valid_lens)
        Y = self.addnorm1(X, X2)
        # Encoder-decoder attention. Shape of `enc_outputs` is: (`batch_size`, `num_steps`, `num_hiddens`)
        Y2 = self.attention2(Y, enc_outputs, enc_outputs, enc_valid_lens)
        Z = self.addnorm2(Y, Y2)
        return self.addnorm3(Z, self.ffn(Z)), state
```

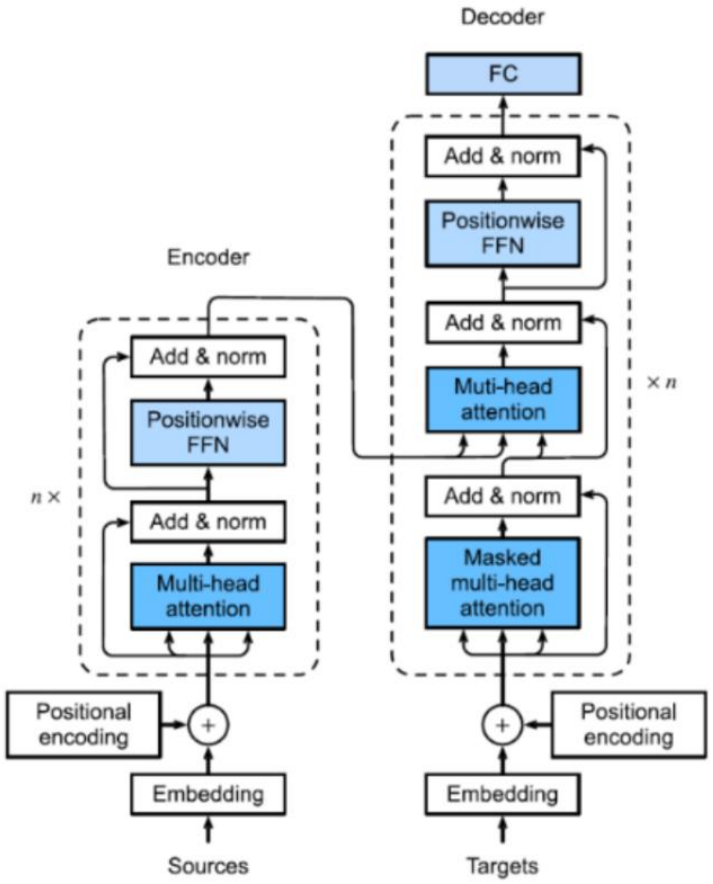
```python
class TransformerDecoder(AttentionDecoder):
    def __init__(self, vocab_size, key_size, query_size, value_size, num_hiddens, norm_shape, ffn_num_input, ffn_num_hiddens, num_heads, num_layers, dropout, **kwargs):
        super(TransformerDecoder, self).__init__(**kwargs)
        self.num_hiddens = num_hiddens
        self.num_layers = num_layers
        self.embedding = nn.Embedding(vocab_size, num_hiddens)
        self.pos_encoding = PositionalEncoding(num_hiddens, dropout)
        self.blks = nn.Sequential()
        for i in range(num_layers):
            self.blks.add_module("block"+str(i),
                DecoderBlock(key_size, query_size, value_size, num_hiddens, norm_shape, ffn_num_input, ffn_num_hiddens, num_heads, dropout, i))
        self.dense = nn.Linear(num_hiddens, vocab_size)

    def init_state(self, enc_outputs, enc_valid_lens, *args):
        return [enc_outputs, enc_valid_lens, [None] * self.num_layers]

    def forward(self, X, state):
        X = self.pos_encoding(self.embedding(X) * math.sqrt(self.num_hiddens))
        self._attention_weights = [[None] * len(self.blks) for _ in range (2)]
        for i, blk in enumerate(self.blks):
            X, state = blk(X, state)
            # Decoder self-attention weights
            self._attention_weights[0][i] = blk.attention1.attention.attention_weights
            # Encoder-decoder attention weights
            self._attention_weights[1][i] = blk.attention2.attention.attention_weights
        return self.dense(X), state

    @property
    def attention_weights(self):
        return self._attention_weights
```

Create the Transformer model in code



putting everything together:

```
encoder = TransformerEncoder(
    len(src_vocab), key_size, query_size, value_size, num_hiddens,
    norm_shape, ffn_num_input, ffn_num_hiddens, num_heads,
    num_layers, dropout)
decoder = TransformerDecoder(
    len(tgt_vocab), key_size, query_size, value_size, num_hiddens,
    norm_shape, ffn_num_input, ffn_num_hiddens, num_heads,
    num_layers, dropout)
net = EncoderDecoder(encoder, decoder)
```

just 12 classes and 4 functions to copy-paste ☺

Exercises

Laboratory 11 – Exercises

1. Generate the following dataset:

$$y_i = -4\cos x_i + 3x_i + \epsilon,$$

where ϵ obeys a normal distribution with mean 0 and standard deviation 0.08. The number of training/testing examples is 100. Plot the prediction for this regression problem using the Nadaraya-Watson kernel regression non-parametric attention model, and also plot the attention heatmap.

2. Train a Transformer model for machine translation. The dataset is the English-French dataset from the laboratory. Modify the `PositionWiseFFN` class by using 3 hidden layers of 32 units. The dropout probability will be set to 70%. Train the model for 400 epochs and plot the training loss.

Solutions

1. Generate the following dataset:

$$y_i = -4\cos x_i + 3x_i + \epsilon,$$

where ϵ obeys a normal distribution with mean 0 and standard deviation 0.08. The number of training/testing examples is 100. Plot the prediction for this regression problem using the Nadaraya-Watson kernel regression non-parametric attention model, and also plot the attention heatmap.

```python
import torch
from torch import nn
import torch.nn.functional as F
import matplotlib.pyplot as plt
import math
import os
import requests
import zipfile
import collections

# Set random seed for reproducibility
torch.manual_seed(42);
```

```python
# Define the regression function
def f(x):
    return (-4) * torch.cos(x) + 3*x
```

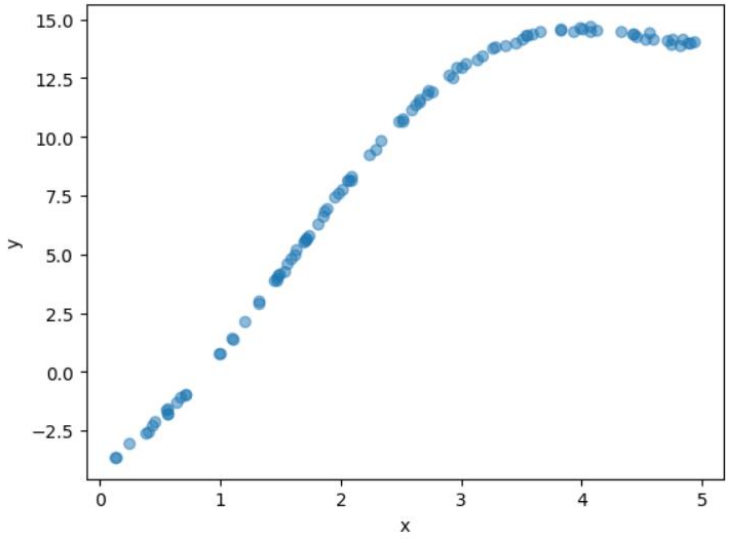
```python
# Generate training data
n_train = 100  # Number of training examples
x_train, _ = torch.sort(torch.rand(n_train) * 5)  # Training inputs in range [0, 5]
y_train = f(x_train) + torch.normal(0, 0.08, (n_train,))  # Training outputs with Gaussian noise

# Generate test data
x_test = torch.arange(0, 5, 0.05)  # Testing inputs from 0 to 5 with step size 0.05
y_truth = f(x_test)  # True outputs for the testing inputs
n_test = len(x_test)  # Number of test examples
n_test
```

```
100
```

I changed this because x_train is in range [0, 5] and we need to generate 100 testing examples

```python
# Define a function to plot training data
def plot_training_data():
    plt.xlabel('x')
    plt.ylabel('y')
    plt.plot(x_train, y_train, 'o', alpha=0.5); # Training data points
```



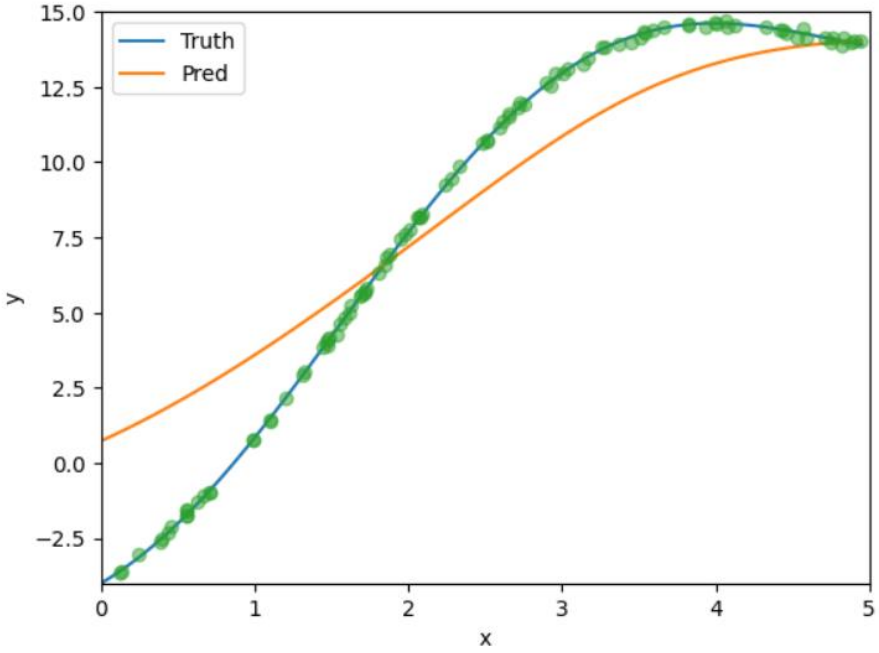
- this is something **optional**, just in case you want to visualize the training data
- it’ s just an adaptation from plot_kernel_reg() function from the lab

```python
# Define a function to plot predictions
def plot_kernel_reg(y_hat):
    plt.plot(x_test, y_truth, label='Truth') # Plot ground truth
    plt.plot(x_test, y_hat, label='Pred') # Plot predictions
    plt.xlabel('x')
    plt.ylabel('y')
    plt.xlim([0, 5])
    plt.ylim([-4, 15])
    plt.legend()
    plt.plot(x_train, y_train, 'o', alpha=0.5); # Training data points
```

I changed only the limits for Ox and Oy axis ☺

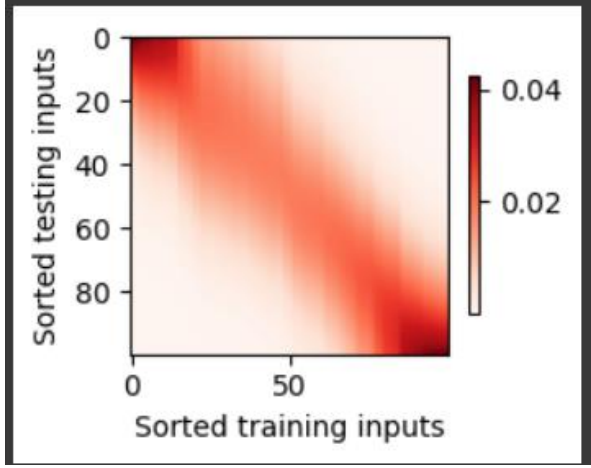
```python
# Shape of `X_repeat` is: (`n_test`, `n_train`), where each row contains the
# same testing inputs (i.e., same queries)
X_repeat = x_test.repeat_interleave(n_train).reshape((-1, n_train))
# Note that `x_train` contains the keys. Shape of `attention_weights` is:
# (`n_test`, `n_train`), where each row contains attention weights to be
# assigned among the values (`y_train`), given each query
attention_weights = F.softmax(-(X_repeat - x_train)**2 / 2, dim=1)
# Each element of `y_hat` is a weighted average of values, where weights are attention weights
y_hat = torch.matmul(attention_weights, y_train)
plot_kernel_reg(y_hat) # Plot predictions
```

copy-paste from lab ☺



```python
def show_heatmaps(matrices, xlabel, ylabel, titles=None, figsize=(2.5, 2.5),
                  cmap='Reds'):
    """Show heatmaps of matrices."""
    num_rows, num_cols = matrices.shape[0], matrices.shape[1]
    fig, axes = plt.subplots(num_rows, num_cols, figsize=figsize,
                             sharex=True, sharey=True, squeeze=False)
    for i, (row_axes, row_matrices) in enumerate(zip(axes, matrices)):
        for j, (ax, matrix) in enumerate(zip(row_axes, row_matrices)):
            pcm = ax.imshow(matrix.detach().numpy(), cmap=cmap)
            if i == num_rows - 1:
                ax.set_xlabel(xlabel)
            if j == 0:
                ax.set_ylabel(ylabel)
            if titles:
                ax.set_title(titles[j])
    fig.colorbar(pcm, ax=axes, shrink=0.6);
```

```python
show_heatmaps(attention_weights.unsqueeze(0).unsqueeze(0),
              xlabel='Sorted training inputs',
              ylabel='Sorted testing inputs')
```



2. Train a Transformer model for machine translation. The dataset is the English-French dataset from the laboratory. Modify the `PositionWiseFFN` class by using 3 hidden layers of 32 units. The dropout probability will be set to 70%. Train the model for 400 epochs and plot the training loss.

- create the Transformer model (a lot of classes and functions):

```python
def sequence_mask(X, valid_len, value=0):
    """Mask irrelevant entries in sequences."""
    maxlen = X.size(1)
    mask = torch.arange((maxlen), dtype=torch.float32,
                        device=X.device)[None, :] < valid_len[:, None]
    X[~mask] = value
    return X

def masked_softmax(X, valid_lens):
    """Perform softmax operation by masking elements on the last axis."""
    # `X`: 3D tensor, `valid_lens`: 1D or 2D tensor
    if valid_lens is None:
        return nn.functional.softmax(X, dim=-1)
    else:
        shape = X.shape
        if valid_lens.dim() == 1:
            valid_lens = torch.repeat_interleave(valid_lens, shape[1])
        else:
            valid_lens = valid_lens.reshape(-1)
        # On the last axis, replace masked elements with a very large negative
        # value, whose exponentiation outputs 0
        X = sequence_mask(X.reshape(-1, shape[-1]), valid_lens,
                          value=-1e6)
        return F.softmax(X.reshape(shape), dim=-1)
```

```python
class DotProductAttention(nn.Module):
    """Scaled dot product attention."""
    def __init__(self, dropout, **kwargs):
        super(DotProductAttention, self).__init__(**kwargs)
        self.dropout = nn.Dropout(dropout)

    # Shape of `queries` is: (`batch_size`, no. of queries, `d`)
    # Shape of `keys` is: (`batch_size`, no. of key-value pairs, `d`)
    # Shape of `values` is: (`batch_size`, no. of key-value pairs, value
    # dimension)
    # Shape of `valid_lens`: (`batch_size`,) or (`batch_size`, no. of queries)
    def forward(self, queries, keys, values, valid_lens=None):
        d = queries.shape[-1]
        # Use `keys.transpose(1,2)` to swap the last two dimensions of `keys`
        scores = torch.bmm(queries, keys.transpose(1,2)) / math.sqrt(d)
        self.attention_weights = masked_softmax(scores, valid_lens)
        return torch.bmm(self.dropout(self.attention_weights), values)
```

```python
def transpose_qkv(X, num_heads):
    """Transposition for parallel computation of multiple attention heads."""
    # Shape of input `X` is:
    # (`batch_size`, no. of queries or key-value pairs, `num_hiddens`).
    # Shape of output `X` is:
    # (`batch_size`, no. of queries or key-value pairs, `num_heads`,
    # `num_hiddens` / `num_heads`)
    X = X.reshape(X.shape[0], X.shape[1], num_heads, -1)

    # Shape of output `X` is:
    # (`batch_size`, `num_heads`, no. of queries or key-value pairs,
    # `num_hiddens` / `num_heads`)
    X = X.permute(0, 2, 1, 3)

    # Shape of `output` is:
    # (`batch_size` * `num_heads`, no. of queries or key-value pairs,
    # `num_hiddens` / `num_heads`)
    return X.reshape(-1, X.shape[2], X.shape[3])

def transpose_output(X, num_heads):
    """Reverse the operation of `transpose_qkv`."""
    X = X.reshape(-1, num_heads, X.shape[1], X.shape[2])
    X = X.permute(0, 2, 1, 3)
    return X.reshape(X.shape[0], X.shape[1], -1)
```

```python
class MultiHeadAttention(nn.Module):
    """Multi-head attention."""
    def __init__(self, key_size, query_size, value_size, num_hiddens,
                 num_heads, dropout, bias=False, **kwargs):
        super(MultiHeadAttention, self).__init__(**kwargs)
        self.num_heads = num_heads
        self.attention = DotProductAttention(dropout)
        self.W_q = nn.Linear(query_size, num_hiddens, bias=bias)
        self.W_k = nn.Linear(key_size, num_hiddens, bias=bias)
        self.W_v = nn.Linear(value_size, num_hiddens, bias=bias)
        self.W_o = nn.Linear(num_hiddens, num_hiddens, bias=bias)

    def forward(self, queries, keys, values, valid_lens):
        # Shape of `queries`, `keys`, or `values` is:
        # (`batch_size`, no. of queries or key-value pairs, `num_hiddens`)
        # Shape of `valid_lens` is:
        # (`batch_size`,) or (`batch_size`, no. of queries)
        # After transposing, shape of output `queries`, `keys`, or `values` is:
        # (`batch_size` * `num_heads`, no. of queries or key-value pairs,
        # `num_hiddens` / `num_heads`)
        queries = transpose_qkv(self.W_q(queries), self.num_heads)
        keys = transpose_qkv(self.W_k(keys), self.num_heads)
        values = transpose_qkv(self.W_v(values), self.num_heads)

        if valid_lens is not None:
            # On axis 0, copy the first item (scalar or vector) for
            # `num_heads` times, then copy the next item, and so on
            valid_lens = torch.repeat_interleave(
                valid_lens, repeats=self.num_heads, dim=0)

        # Shape of `output` is: (`batch_size` * `num_heads`, no. of queries,
        # `num_hiddens` / `num_heads`)
        output = self.attention(queries, keys, values, valid_lens)

        # Shape of `output_concat` is:
        # (`batch_size`, no. of queries, `num_hiddens`)
        output_concat = transpose_output(output, self.num_heads)
        return self.W_o(output_concat)
```

```python
class PositionalEncoding(nn.Module):
    """Positional encoding."""
    def __init__(self, num_hiddens, dropout, max_len=1000):
        super(PositionalEncoding, self).__init__()
        self.dropout = nn.Dropout(dropout)
        # Create a long enough `P`
        self.P = torch.zeros((1, max_len, num_hiddens))
        X = torch.arange(max_len, dtype=torch.float32).reshape(
            -1, 1) / torch.pow(10000, torch.arange(
            0, num_hiddens, 2, dtype=torch.float32) / num_hiddens)
        self.P[:, :, 0::2] = torch.sin(X)
        self.P[:, :, 1::2] = torch.cos(X)

    def forward(self, X):
        X = X + self.P[:, :X.shape[1], :].to(X.device)
        return self.dropout(X)
```

```python
class AddNorm(nn.Module):
    """Residual connection followed by layer normalization."""
    def __init__(self, normalized_shape, dropout, **kwargs):
        super(AddNorm, self).__init__(**kwargs)
        self.dropout = nn.Dropout(dropout)
        self.ln = nn.LayerNorm(normalized_shape)

    def forward(self, X, Y):
        return self.ln(self.dropout(Y) + X)
```

```python
# Class modified
class PositionWiseFFN(nn.Module):
    """Position-wise feed-forward network."""
    def __init__(self, ffn_num_input, ffn_num_hiddens, ffn_num_outputs,
                 **kwargs):
        super(PositionWiseFFN, self).__init__(**kwargs)
        self.dense1 = nn.Linear(ffn_num_input, ffn_num_hiddens)
        self.dense2 = nn.Linear(ffn_num_hiddens, ffn_num_hiddens)
        self.dense3 = nn.Linear(ffn_num_hiddens, ffn_num_hiddens)
        self.relu = nn.ReLU()
        self.dense4 = nn.Linear(ffn_num_hiddens, ffn_num_outputs)


    def forward(self, X):
        return self.dense4(self.relu(self.dense3(self.relu(self.dense2(self.relu(self.dense1(X)))))))
```

Modify the `PositionWiseFFN` class by using 3 hidden layers of 32 units. The dropout probability will be set to 70%.

```python
class Encoder(nn.Module):
    """The base encoder interface for the encoder-decoder architecture."""
    def __init__(self, **kwargs):
        super(Encoder, self).__init__(**kwargs)

    def forward(self, X, *args):
        raise NotImplementedError

class EncoderBlock(nn.Module):
    """Transformer encoder block."""
    def __init__(self, key_size, query_size, value_size, num_hiddens,
                 norm_shape, ffn_num_input, ffn_num_hiddens, num_heads,
                 dropout, use_bias=False, **kwargs):
        super(EncoderBlock, self).__init__(**kwargs)
        self.attention = MultiHeadAttention(
            key_size, query_size, value_size, num_hiddens, num_heads, dropout,
            use_bias)
        self.addnorm1 = AddNorm(norm_shape, dropout)
        self.ffn = PositionWiseFFN(
            ffn_num_input, ffn_num_hiddens, num_hiddens)
        self.addnorm2 = AddNorm(norm_shape, dropout)

    def forward(self, X, valid_lens):
        Y = self.addnorm1(X, self.attention(X, X, X, valid_lens))
        return self.addnorm2(Y, self.ffn(Y))
```

```python
class TransformerEncoder(Encoder):
    """Transformer encoder."""
    def __init__(self, vocab_size, key_size, query_size, value_size,
                 num_hiddens, norm_shape, ffn_num_input, ffn_num_hiddens,
                 num_heads, num_layers, dropout, use_bias=False, **kwargs):
        super(TransformerEncoder, self).__init__(**kwargs)
        self.num_hiddens = num_hiddens
        self.embedding = nn.Embedding(vocab_size, num_hiddens)
        self.pos_encoding = PositionalEncoding(num_hiddens, dropout)
        self.blks = nn.Sequential()
        for i in range(num_layers):
            self.blks.add_module("block"+str(i),
                EncoderBlock(key_size, query_size, value_size, num_hiddens,
                             norm_shape, ffn_num_input, ffn_num_hiddens,
                             num_heads, dropout, use_bias))

    def forward(self, X, valid_lens, *args):
        # Since positional encoding values are between -1 and 1, the embedding
        # values are multiplied by the square root of the embedding dimension
        # to rescale before they are summed up
        X = self.pos_encoding(self.embedding(X) * math.sqrt(self.num_hiddens))
        self.attention_weights = [None] * len(self.blks)
        for i, blk in enumerate(self.blks):
            X = blk(X, valid_lens)
            self.attention_weights[
                i] = blk.attention.attention.attention_weights
        return X
```

```python
class Decoder(nn.Module):
    """The base decoder interface for the encoder-decoder architecture."""
    def __init__(self, **kwargs):
        super(Decoder, self).__init__(**kwargs)

    def init_state(self, enc_outputs, *args):
        raise NotImplementedError

    def forward(self, X, state):
        raise NotImplementedError

class AttentionDecoder(Decoder):
    """The base attention-based decoder interface."""
    def __init__(self, **kwargs):
        super(AttentionDecoder, self).__init__(**kwargs)

    @property
    def attention_weights(self):
        raise NotImplementedError
```

```python
class TransformerDecoder(AttentionDecoder):
    def __init__(self, vocab_size, key_size, query_size, value_size,
                 num_hiddens, norm_shape, ffn_num_input, ffn_num_hiddens,
                 num_heads, num_layers, dropout, **kwargs):
        super(TransformerDecoder, self).__init__(**kwargs)
        self.num_hiddens = num_hiddens
        self.num_layers = num_layers
        self.embedding = nn.Embedding(vocab_size, num_hiddens)
        self.pos_encoding = PositionalEncoding(num_hiddens, dropout)
        self.blks = nn.Sequential()
        for i in range(num_layers):
            self.blks.add_module("block"+str(i),
                DecoderBlock(key_size, query_size, value_size, num_hiddens,
                             norm_shape, ffn_num_input, ffn_num_hiddens,
                             num_heads, dropout, i))
        self.dense = nn.Linear(num_hiddens, vocab_size)

    def init_state(self, enc_outputs, enc_valid_lens, *args):
        return [enc_outputs, enc_valid_lens, [None] * self.num_layers]

    def forward(self, X, state):
        X = self.pos_encoding(self.embedding(X) * math.sqrt(self.num_hiddens))
        self._attention_weights = [[None] * len(self.blks) for _ in range (2)]
        for i, blk in enumerate(self.blks):
            X, state = blk(X, state)
            # Decoder self-attention weights
            self._attention_weights[0][
                i] = blk.attention1.attention.attention_weights
            # Encoder-decoder attention weights
            self._attention_weights[1][
                i] = blk.attention2.attention.attention_weights
        return self.dense(X), state

    @property
    def attention_weights(self):
        return self._attention_weights
```

```python
class DecoderBlock(nn.Module):
    # The `i`-th block in the decoder
    def __init__(self, key_size, query_size, value_size, num_hiddens,
                 norm_shape, ffn_num_input, ffn_num_hiddens, num_heads,
                 dropout, i, **kwargs):
        super(DecoderBlock, self).__init__(**kwargs)
        self.i = i
        self.attention1 = MultiHeadAttention(
            key_size, query_size, value_size, num_hiddens, num_heads, dropout)
        self.addnorm1 = AddNorm(norm_shape, dropout)
        self.attention2 = MultiHeadAttention(
            key_size, query_size, value_size, num_hiddens, num_heads, dropout)
        self.addnorm2 = AddNorm(norm_shape, dropout)
        self.ffn = PositionWiseFFN(ffn_num_input, ffn_num_hiddens,
                                   num_hiddens)
        self.addnorm3 = AddNorm(norm_shape, dropout)

    def forward(self, X, state):
        enc_outputs, enc_valid_lens = state[0], state[1]
        # During training, all the tokens of any output sequence are processed
        # at the same time, so `state[2][self.i]` is `None` as initialized.
        # When decoding any output sequence token by token during prediction,
        # `state[2][self.i]` contains representations of the decoded output at
        # the `i`-th block up to the current time step
        if state[2][self.i] is None:
            key_values = X
        else:
            key_values = torch.cat((state[2][self.i], X), axis=1)
        state[2][self.i] = key_values
        if self.training:
            batch_size, num_steps, _ = X.shape
            # Shape of `dec_valid_lens` is: (`batch_size`, `num_steps`), where
            # every row is [1, 2, ..., `num_steps`]
            dec_valid_lens = torch.arange(
                1, num_steps + 1, device=X.device).repeat(batch_size, 1)
        else:
            dec_valid_lens = None

        # Self-attention
        X2 = self.attention1(X, key_values, key_values, dec_valid_lens)
        Y = self.addnorm1(X, X2)
        # Encoder-decoder attention. Shape of `enc_outputs` is:
        # (`batch_size`, `num_steps`, `num_hiddens`)
        Y2 = self.attention2(Y, enc_outputs, enc_outputs, enc_valid_lens)
        Z = self.addnorm2(Y, Y2)
        return self.addnorm3(Z, self.ffn(Z)), state
```

```python
class EncoderDecoder(nn.Module):
    """The base class for the encoder-decoder architecture."""
    def __init__(self, encoder, decoder, **kwargs):
        super(EncoderDecoder, self).__init__(**kwargs)
        self.encoder = encoder
        self.decoder = decoder

    def forward(self, enc_X, dec_X, *args):
        enc_outputs = self.encoder(enc_X, *args)
        dec_state = self.decoder.init_state(enc_outputs, *args)
        return self.decoder(dec_X, dec_state)
```

```python
def try_gpu(i=0):
    """Return gpu(i) if exists, otherwise return cpu()."""
    if torch.cuda.device_count() >= i + 1:
        return torch.device(f'cuda:{i}')
    return torch.device('cpu')
```

```python
def count_corpus(tokens):
    """Count token frequencies."""
    # Here `tokens` is a 1D list or 2D list
    if len(tokens) == 0 or isinstance(tokens[0], list):
        # Flatten a list of token lists into a list of tokens
        tokens = [token for line in tokens for token in line]
    return collections.Counter(tokens)

def download(url, cache_dir=os.path.join('..', 'data')):
    """Download a file, return the local filename."""
    os.makedirs(cache_dir, exist_ok=True)
    fname = os.path.join(cache_dir, url.split('/')[-1])
    if os.path.exists(fname):
        with open(fname, 'rb') as f:
            while True:
                data = f.read(1048576)
                if not data:
                    break
        return fname
    print(f'Downloading {fname} from {url}...')
    r = requests.get(url, stream=True, verify=True)
    with open(fname, 'wb') as f:
        f.write(r.content)
    return fname
```

```python
class Vocab:
    """Vocabulary for text."""
    def __init__(self, tokens=None, min_freq=0, reserved_tokens=None):
        if tokens is None:
            tokens = []
        if reserved_tokens is None:
            reserved_tokens = []
        # Sort according to frequencies
        counter = count_corpus(tokens)
        self._token_freqs = sorted(counter.items(), key=lambda x: x[1],
                                   reverse=True)
        # The index for the unknown token is 0
        self.idx_to_token = ['<unk>'] + reserved_tokens
        self.token_to_idx = {token: idx
                             for idx, token in enumerate(self.idx_to_token)}
        for token, freq in self._token_freqs:
            if freq < min_freq:
                break
            if token not in self.token_to_idx:
                self.idx_to_token.append(token)
                self.token_to_idx[token] = len(self.idx_to_token) - 1

    def __len__(self):
        return len(self.idx_to_token)

    def __getitem__(self, tokens):
        if not isinstance(tokens, (list, tuple)):
            return self.token_to_idx.get(tokens, self.unk)
        return [self.__getitem__(token) for token in tokens]

    def to_tokens(self, indices):
        if not isinstance(indices, (list, tuple)):
            return self.idx_to_token[indices]
        return [self.idx_to_token[index] for index in indices]

    @property
    def unk(self):  # Index for the unknown token
        return 0

    @property
    def token_freqs(self):  # Token frequencies
        return self._token_freqs
```

```python
def download_extract(url, folder=None):
    """Download and extract a zip file."""
    fname = download(url)
    base_dir = os.path.dirname(fname)
    data_dir, ext = os.path.splitext(fname)
    if ext == '.zip':
        fp = zipfile.ZipFile(fname, 'r')
    else:
        assert False, 'Only zip files can be extracted.'
    fp.extractall(base_dir)
    return os.path.join(base_dir, folder) if folder else data_dir

def read_data_nmt():
    """Load the English-French dataset."""
    data_dir = download_extract('http://d2l-data.s3-accelerate.amazonaws.com/fra-eng.zip')
    with open(os.path.join(data_dir, 'fra.txt'), 'r') as f:
        return f.read()

def preprocess_nmt(text):
    """Preprocess the English-French dataset."""
    def no_space(char, prev_char):
        return char in set(',.!?') and prev_char != ' '

    # Replace non-breaking space with space, and convert uppercase letters to
    # lowercase ones
    text = text.replace('\u202f', ' ').replace('\xa0', ' ').lower()
    # Insert space between words and punctuation marks
    out = [' ' + char if i > 0 and no_space(char, text[i - 1]) else char
           for i, char in enumerate(text)]
    return ''.join(out)
```

```python
def tokenize_nmt(text, num_examples=None):
    """Tokenize the English-French dataset."""
    source, target = [], []
    for i, line in enumerate(text.split('\n')):
        if num_examples and i > num_examples:
            break
        parts = line.split('\t')
        if len(parts) == 2:
            source.append(parts[0].split(' '))
            target.append(parts[1].split(' '))
    return source, target

def truncate_pad(line, num_steps, padding_token):
    """Truncate or pad sequences."""
    if len(line) > num_steps:
        return line[:num_steps]  # Truncate
    return line + [padding_token] * (num_steps - len(line))  # Pad

def build_array_nmt(lines, vocab, num_steps):
    """Transform text sequences of machine translation into mini-batches."""
    lines = [vocab[l] for l in lines]
    lines = [l + [vocab['<eos>']] for l in lines]
    array = torch.tensor([truncate_pad(
        l, num_steps, vocab['<pad>']) for l in lines])
    valid_len = (array != vocab['<pad>']).type(torch.int32).sum(1)
    return array, valid_len

def load_array(data_arrays, batch_size, is_train=True):
    """Construct a PyTorch data iterator."""
    dataset = torch.utils.data.TensorDataset(*data_arrays)
    return torch.utils.data.DataLoader(dataset, batch_size, shuffle=is_train)

def load_data_nmt(batch_size, num_steps, num_examples=600):
    """Return the iterator and the vocabularies of the translation dataset."""
    text = preprocess_nmt(read_data_nmt())
    source, target = tokenize_nmt(text, num_examples)
    src_vocab = Vocab(source, min_freq=2,
                      reserved_tokens=['<pad>', '<bos>', '<eos>'])
    tgt_vocab = Vocab(target, min_freq=2,
                      reserved_tokens=['<pad>', '<bos>', '<eos>'])
    src_array, src_valid_len = build_array_nmt(source, src_vocab, num_steps)
    tgt_array, tgt_valid_len = build_array_nmt(target, tgt_vocab, num_steps)
    data_arrays = (src_array, src_valid_len, tgt_array, tgt_valid_len)
    data_iter = load_array(data_arrays, batch_size)
    return data_iter, src_vocab, tgt_vocab
```

```python
class MaskedSoftmaxCELoss(nn.CrossEntropyLoss):
    """The softmax cross-entropy loss with masks."""
    # `pred` shape: (`batch_size`, `num_steps`, `vocab_size`)
    # `label` shape: (`batch_size`, `num_steps`)
    # `valid_len` shape: (`batch_size`,)
    def forward(self, pred, label, valid_len):
        weights = torch.ones_like(label)
        weights = sequence_mask(weights, valid_len)
        self.reduction='none'
        unweighted_loss = super(MaskedSoftmaxCELoss, self).forward(
            pred.permute(0, 2, 1), label)
        weighted_loss = (unweighted_loss * weights).mean(dim=1)
        return weighted_loss

def grad_clipping(net, theta):
    """Clip the gradient."""
    params = [p for p in net.parameters() if p.requires_grad]
    norm = torch.sqrt(sum(torch.sum((p.grad ** 2)) for p in params))
    if norm > theta:
        for param in params:
            param.grad[:] *= theta / norm
```

```python
def train_seq2seq(net, data_iter, lr, num_epochs, tgt_vocab, device):
    """Train a model for sequence to sequence."""
    def xavier_init_weights(m):
        if type(m) == nn.Linear:
            nn.init.xavier_uniform_(m.weight)
        if type(m) == nn.GRU:
            for param in m._flat_weights_names:
                if "weight" in param:
                    nn.init.xavier_uniform_(m._parameters[param])
    net.apply(xavier_init_weights)
    net.to(device)
    optimizer = torch.optim.Adam(net.parameters(), lr=lr)
    loss = MaskedSoftmaxCELoss()
    net.train()
    train_loss_all = []
    for epoch in range(num_epochs):
        # Sum of training loss, no. of tokens
        total_loss = 0
        total_tokens = 0
        for batch in data_iter:
            optimizer.zero_grad()
            X, X_valid_len, Y, Y_valid_len = [x.to(device) for x in batch]
            bos = torch.tensor([tgt_vocab['<bos>']] * Y.shape[0],
                               device=device).reshape(-1, 1)
            dec_input = torch.cat([bos, Y[:, :-1]], 1)  # Teacher forcing
            Y_hat, _ = net(X, dec_input, X_valid_len)
            l = loss(Y_hat, Y, Y_valid_len)
            l.sum().backward()  # Make the loss scalar for `backward`
            grad_clipping(net, 1)
            num_tokens = Y_valid_len.sum()
            optimizer.step()
            with torch.no_grad():
                total_loss += l.sum()
                total_tokens += num_tokens
        if (epoch + 1) % 10 == 0:
            train_loss_all.append(total_loss / total_tokens)
    print(f'loss {total_loss / total_tokens:.3f}, device {str(device)}')

    return train_loss_all
```

```python
num_hiddens, num_layers, dropout, batch_size, num_steps = 32, 2, 0.7, 64, 10
lr, num_epochs, device = 0.005, 400, try_gpu()
ffn_num_input, ffn_num_hiddens, num_heads = 32, 64, 4
key_size, query_size, value_size = 32, 32, 32
norm_shape = [32]

train_iter, src_vocab, tgt_vocab = load_data_nmt(batch_size, num_steps)

encoder = TransformerEncoder(
    len(src_vocab), key_size, query_size, value_size, num_hiddens,
    norm_shape, ffn_num_input, ffn_num_hiddens, num_heads,
    num_layers, dropout)
decoder = TransformerDecoder(
    len(tgt_vocab), key_size, query_size, value_size, num_hiddens,
    norm_shape, ffn_num_input, ffn_num_hiddens, num_heads,
    num_layers, dropout)
net = EncoderDecoder(encoder, decoder)
train_loss_all = train_seq2seq(net, train_iter, lr, num_epochs, tgt_vocab, device)
```

```
Downloading ../data/fra-eng.zip from http://d2l-data.s3-accelerate.amazonaws.com/fra-eng.zip...
loss 0.185, device cpu
```

```python
def plot_loss(train_loss_all):
    epochs = range(10, len(train_loss_all * 10) + 1, 10)
    plt.plot(epochs, train_loss_all, 'b', label='Train loss')
    plt.title('Training loss')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()
    plt.show()

plot_loss(train_loss_all)
```

only 15 classes and 18 functions to copy-paste ☺

Thank you

for your $\mathrm{softmax}(\frac{QK^T}{\sqrt{d_k}})V$!